are usually treated with a 7- to 14-day antibiotic course. A study conducted in male veterans found shorter-course treatments for UTI (< 7 days) to be as effective compared to longer duration of treatment (> 7 days). Another study compared 7- and 14-day courses for patients with febrile UTI. In men, 7 days of antibiotics was inferior to 14 days in achieving short-term (10–18 days posttreatment) clinical cure. However, the long-term clinical cure rates were similar indicating no difference in outcomes at 70 to 84 days posttreatment. Therefore, for most older men with acute simple cystitis 7 days of antibiotic treatment should be adequate if they respond promptly to antibiotics (within 72 hours).

Management of complicated UTI (infection extending beyond the bladder including pyelonephritis) varies depend on the severity of the illness. Although age more than 60 years has been described to be a relative indication for hospitalization in patients with acute pyelonephritis, outpatient management in those with mild to moderate illness, adequate social support, and clinical stability is likely to be as safe and as effective compared to hospitalization. Older adults with complicated UTI (including pyelonephritis) who present with severe symptoms (eg, persistent vomiting), suspected sepsis, uncertain diagnosis, evidence of urinary tract obstruction, and have significant frailty should be hospitalized. FQs are the only oral antibiotic recommended for empiric treatment of complicated UTI. However, increasing resistance of FQ and concerns of side effects limit the use of this drug in many clinical settings. Experts do not recommend empiric FQ use in areas with local resistance noted to be higher than 10%. Similarly, FQ use should also be avoided if patient at high risk for infection with MDROs or adverse events are secondary to its use. If FQ is being used for treatment of complicated UTI in areas with more than 10% resistance, then the patient should receive an initial dose of long-acting parenteral agent (eg, ceftriaxone, ertapenem, etc) before starting FQ. Similarly, if using any other oral antibiotic (eg, TMP-SMX) for empiric treatment of complicated UTI, patients should receive one dose of long-acting parenteral agent before starting the oral antibiotic. Clinicians may consider continuing parenteral treatment and wait for the culture and susceptibility results before switching to oral antibiotics especially if the patient is at higher risk of infection secondary to MDROs or severe illness. Oral beta-lactam agents are usually less effective than FQ and TMP-SMX for treatment of complicated UTI including acute pyelonephritis and should be avoided when other treatment options are available. Nitrofurantoin and fosfomycin should not be used for complicated UTI because of inadequate renal penetration. When choosing parenteral agent for the initial dose, aminoglycosides (such as gentamicin or tobramycin) are usually avoided because they are more toxic than other available parenteral antibiotics (eg, ceftriaxone, ertapenem, etc), although the risk of ototoxicity and nephrotoxicity with aminoglycoside therapy is very low when therapy duration is limited to 48 to 72 hours (see **Table 106-6**).

Pharmacologic therapies for postacute and LTC residents In general, pharmacologic therapies for postacute and LTC residents are usually similar to what have been described above for community-dwelling older adults. Some of the factors that usually influence decision-making for treatment of UTI may need additional attention in postacute and LTC settings. First, antibiotic resistance is generally much higher in these settings compared to the community population. Thus, it is imperative that clinicians become familiar with local resistance rates and guide empiric antibiotic regimens based on resistance patterns. Urine culture results should always help adjust antibiotic choices. Second, alterations in antimicrobial pharmacokinetics such as increased volume of distribution and decreased renal clearance are often observed in older adults; thus, providers should be cognizant of a patient's glomerular filtration rate (GFR) when dosing antimicrobials. Third, older adults in these settings often must manage polypharmacy and drug–drug interactions should always be evaluated. For instance, TMP/SMX, along with many other antibiotics, has significant interactions with anticoagulants and thus, close international normalized ratio (INR) monitoring is recommended in patients receiving warfarin. In most cases, warfarin dosages will need to be decreased by at least 50% through the entire duration of antimicrobial therapy.

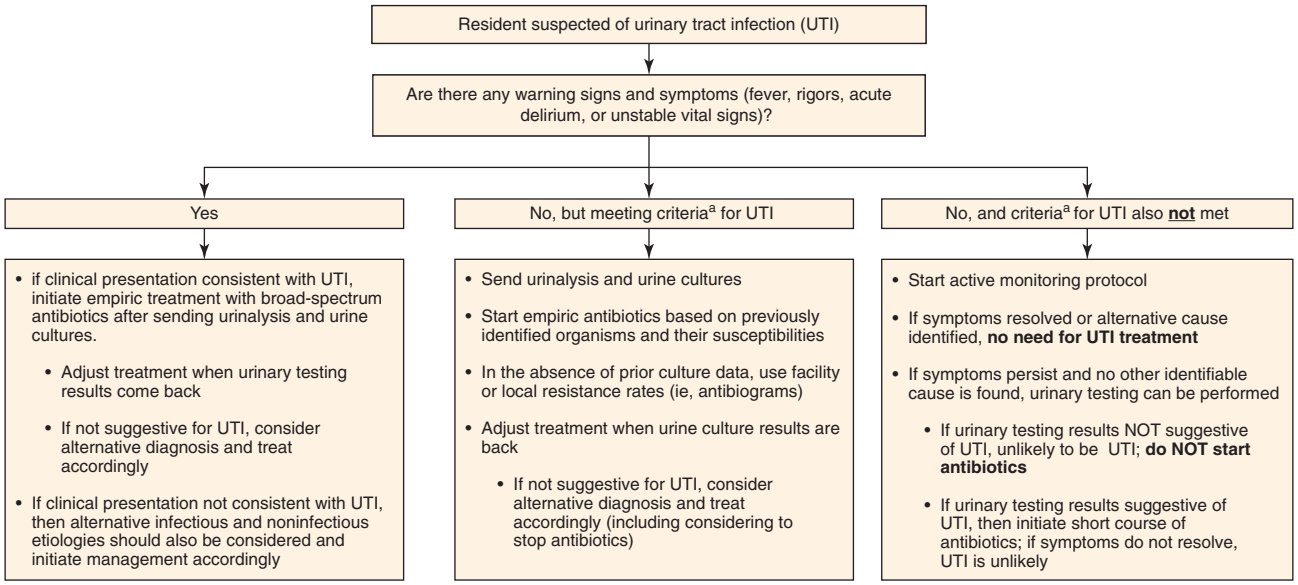
Finally, establishing a diagnosis of UTI is very challenging in frail and medically complex residents of postacute and long-term settings, some of whom may not be able to verbalize any symptoms. In order to avoid initiating unnecessary antibiotic treatment, AMDA's consensus statement recommends that clinicians in PALTC setting use standard clinical algorithms to guide the diagnosis and decision to initiate antibiotics for residents with suspected UTI.

Several consensus-based criteria have been published to assist with diagnosis and treatment of suspected UTI in residents of PALTC settings (see **Table 106-4**). Loeb minimum criteria describe the clinical signs and symptoms that should be present prior to initiating antibiotics in nursing home residents suspected of having infection. AHRQ also developed a set of criteria, which were incorporated into a standard communication and decision aid tool. The UTI SBAR tool developed by AHRQ aims to assist in better assessment of resident signs and symptoms by the nursing home staff, effective communication of those findings with prescribing clinicians, and differentiating between when a resident needs to be started on antibiotic versus on an active monitoring protocol. International Delphi Consensus decision tool consists of separate clinical algorithms for management of suspected UTI in frail older adults with and without indwelling catheters. The decision tool describes which specific and nonspecific signs and symptoms are associated with UTI and when the antibiotic prescribing is justified. The IOU consensus statement also used the Delphi procedure to provide guidance on which clinical signs and symptoms to consider when making decision on empiric treatment of uncomplicated cystitis. There has

been no head-to-head comparison for the clinical impact of the above-mentioned consensus-based criteria. Therefore, the AMDA UTI consensus statement has suggested that PALTC settings should choose one of the consensus-based criteria that seems most closely aligned with their current practices to facilitate the implementation.

Older adults including residents of postacute and LTC settings may manifest acute illness with atypical signs and symptoms that can also be nonlocalizing. Clinicians should perform careful evaluation to differentiate between infectious and noninfectious etiologies for change of condition in residents of PALTC settings. SHEA has published an expert guidance that outlines those signs and symptoms that should and should not prompt further evaluation for infection in these settings. According to this guidance, clinicians should evaluate residents with fever, hypothermia, hypotension, hyperglycemia, and/or unequivocal delirium for an infection while considering the possibility of noninfectious causes for these symptoms. The guidance recommends against further evaluation for infection in residents who present with a fall, functional decline, or anorexia. Furthermore, it recommends using Confusion Assessment Method (CAM) to identify the presence of delirium in a resident of a nursing home with behavioral/mental status changes. Delirium in residents of PALTC settings may be precipitated by a number of infectious and noninfectious conditions, including medications and metabolic disorders. If delirium has been excluded, further evaluation for infection is not necessary unless additional, more specific signs and symptoms are present. The AMDA UTI consensus statement also recommends against initiating empiric treatment for UTI when clinical criteria for UTI are not met and there are no warning signs (ie, evidence of systemic infection, such as fever, rigors/chills, unstable vitals, or new-onset, clear-cut delirium). In those situations, if clinicians are still concerned for UTI, the AMDA consensus statement recommends responding to this diagnostic uncertainty with an "active monitoring" protocol. If the resident condition improves with the supportive care (including hydration), that will obviate the need for treating suspected UTI. The approach to management of suspected UTI in PALTC settings is summarized in **Figure 106-2**.

Shorter length of therapy, similar to what is described for community-dwelling older adult, is also sufficient for treatment of UTI in residents of PALTC settings. The recently published AMDA consensus statement recommended fewer than 7-day antibiotic course to PALTC residents with acute simple cystitis who are not severely ill and not at high risk for developing complications. For those residents who are at higher risk of treatment failure or who have complicated UTI, the length of antibiotic is recommended to be based on the severity of the illness and response to treatment. For most of these residents, 7 days of antibiotic treatment should be adequate if they respond promptly to antibiotics (within 72 hours). Longer (10–14 days) durations are reasonable for residents with severe illness, such as bacteremia, or a delayed response to treatment. The choice of antibiotics also impacts the duration of treatment. For example, when treating complicated UTI, FQs are usually prescribed for 5 to 7 days, TMP-SMX for 7 to 14 days, and oral beta-lactams for 10 to 14 days.



[a]Several consensus-based criteria have been published to assist with diagnosis and treatment of suspected UTI (see text). Postacute and long-term care settings may choose a consensus-based criteria that seem most closely aligned with their current practices to facilitate the implementation.

FIGURE 106-2. Approach to management of suspected UTI in postacute and long-term care settings.

PREVENTION

Cranberry Formulations

Cranberry juice has been used as a home remedy for prevention and treatment of UTI for decades, although the evidence to support this intervention as a prevention strategy is still debated. Cranberry proanthocyanidin (PAC) is the active ingredient in cranberry that is thought to inhibit adherence of P-fimbriated *E coli* to uroepithelial cell receptors. However, a Cochrane review evaluating use of cranberry containing products (ie, juice, tablets, or capsules) did not find significant evidence that any of these products prevented UTI in women compared to placebo, water, or no treatment. Another study examining cranberry capsules containing 36 to 108 mg of PAC found a trend toward a decrease in bacteriuria plus pyuria in LTC residents, suggesting formulations with higher doses of PAC may be beneficial. A systematic review and meta-analysis of only RCTs did conclude that cranberry products reduced UTI recurrence in women with at least three UTIs in a 1-year period (risk ratio [RR] 0.53; 95% CI, 0.33–0.83). However, because of the substantial heterogeneity across trials, the authors mention that this conclusion should be interpreted with great caution. A more recent systematic review and meta-analysis of seven randomized trials came to similar conclusion. They found cranberry reduced the risk of UTI by 26% (pooled RR: 0.74; 95% CI, 0.55–0.98; I^2 = 54%) in healthy women at risk for UTI. However, most of the studies had small sample size and there were significant limitations due to which the authors noted that larger high-quality studies are needed to confirm these findings. Given the issues of volume and glucose load for a cranberry juice supplement and size and cost of cranberry capsules, further investigation of cranberry supplements in older adults is warranted prior to recommending either product as a prevention strategy. Even though more recent urology guidelines stated that clinicians may offer cranberry prophylaxis for recurrent UTI (while acknowledging the limitations), AMDA UTI consensus statement for PALTC setting concluded that current evidence does not support the use of cranberry products for the prevention of UTI.

Vaginal Estrogen

In older adults, declining estrogen levels are thought to lead to declining levels of lactobacilli in the vaginal tract and thus, increase colonization of the perineal area with organisms such as *E coli*. Topical estrogen appears to be a safe and potentially effective method for prevention of UTI in postmenopausal women. A systematic review and meta-analysis evaluated the evidence regarding use of vaginal estrogens as a prevention strategy for recurrent UTI in postmenopausal women. Both RCTs included in this study found that vaginal estrogen (vaginal cream containing 0.5 mg of estriol or vaginal ring with 2 mg estradiol) significantly reduced the risk of UTI in postmenopausal women (RR 0.25; 95% CI, 0.13–0.50; and RR 0.64; 95% CI, 0.47–0.86, respectively). However, their pooled effect was not significant (RR 0.42; 95% CI, 0.16–1.10). In one of the study that described significant decrease in incidence of UTI in the vaginal estriol group as compared to placebo group (0.5 vs 5.9 episodes per-patient year, $p < 0.001$), reported side effects of vaginal estrogen use included vaginal irritation, burning or itching. However, the side effects were described as mild and self-limited. As benefits of vaginal estrogens appear to outweigh potential minor side effect concerns most experts including recent urology guidelines and AMDA UTI consensus statement for PALTC settings recommend considering vaginal estrogen therapy for prevention of recurrent UTI in older women. Vaginal estrogen has not been demonstrated to increase the risk of estrogen-dependent cancer (eg, breast cancer). However, clinicians are recommended to coordinate with patients' oncologists when considering vaginal estrogen therapy for those who either have personal history or are at high risk for estrogen-dependent cancer.

Suppressive Antibiotics

In general, prolonged suppressive antimicrobial therapy for prevention of UTI in older adults should be avoided. It may be considered when there is persistent or recurrent morbidity attributable to UTI. Such circumstances are uncommon, but might include recurrent invasive infection in the presence of an underlying genitourinary abnormality that cannot be corrected, persistent infected stones where suppressive therapy may prevent further stone enlargement, frequent symptomatic recurrences from a prostatic source, or with chronic renal failure. Antimicrobial selection for suppressive therapy is individualized, based on susceptibilities of the infecting organism and tolerance. Deciding whether to use antibiotic prophylaxis must be determined based on the frequency of UTI, severity of symptoms, and possibility of adverse effects of chronic antimicrobial use such as development of *Clostridioides difficile* colitis, selection of multidrug-resistant organisms and various other adverse drug reactions. Patients should also be made aware of the risks and benefits for choosing suppressive antibiotics for prevention of UTI. For many older adults, these risks are likely to outweigh any benefit and suppressive therapy is seldom indicated. More recently, a matched cohort study compared older adults with UTI receiving antibiotic prophylaxis (defined as antibiotic treatment for at least 30 days) that was started within 30 days after a positive urine culture to those who did not receive prophylaxis (but received the treatment). Antibiotic prophylaxis recipients were at higher risk of emergency department visits or hospitalizations for UTI, sepsis or bloodstream infection (HR, 1.33; 95% CI, 1.12–1.57), acquisition of antibiotic resistance to any urinary antibiotics (HR, 1.31; 95% CI, 1.18–1.44), developing *C difficile* infections (HR, 1.56; 95% CI, 1.05–2.23), and experiencing general medical adverse events (HR, 1.62; 95% CI, 1.11–2.29). Recent urology guidelines state that clinicians may prescribe antibiotic prophylaxis to decrease

the risk of future UTI but emphasize that all antibiotics have potential risk and the decision to prescribe should follow the discussion of the risks, benefits, and alternatives with the patients. Recent AMDA UTI consensus statement acknowledges that antibiotic prophylaxis may reduce the risk of recurrent uncomplicated cystitis but points out that the potential harms associated with long-term use, coupled with the prevalence of multidrug-resistant organisms among PALTC residents, argue against long-term antibiotic prophylaxis in these settings.

Other Antimicrobial Sparing Agents

In addition to the cranberry products and vaginal estrogens, several other agents have been used in clinical practice for prevention of UTI but their benefits remain uncertain. These agents include methenamine salts, probiotics, ascorbic acid, and D-Mannose. A structured review published in 2019 evaluated the available evidence for the use of several nonantibiotic agents to prevent UTI in women 45 years or older concluded that there is a lack of evidence to make recommendations for or against the use of ascorbic acid, cranberry juice, cranberry capsules with high proanthocyanidin content, D-mannose, *Lactobacillus*, and methenamine hippurate. Further studies are needed to clarify the roles of these antimicrobial sparing agents in prevention of UTI among older adults.

Prevention of Catheter-Associated UTI

The most effective strategies for prevention of CAUTI are to avoid catheterization and, when an indwelling catheter is indicated, to discontinue the catheter as soon as possible. The Healthcare Infection Control Practices Advisory Committee as part of the Centers for Disease Control (CDC), issued guidelines in 2009, aimed to help providers with the prevention, diagnosis, and management of persons with CAUTI, particularly hospitalized patients and residents of LTCFs. In older adults that require catheter placement, recommendations include insertion of catheters aseptically by trained personnel, use of the smallest diameter catheter as possible, hand washing before and after catheter manipulation, maintenance of a closed catheter system, avoiding irrigation unless anticipating obstruction (eg, after prostatic or bladder surgery), keeping the collecting bag below the bladder, and maintaining good hydration in residents. The guidelines also recommend against changing indwelling catheters or drainage bags at routine, fixed intervals. Instead, it is suggested to change catheters and drainage bags based on clinical indications such as infection, obstruction, or when the closed system is compromised. The catheter should only be replaced if still indicated. Use of antimicrobial-coated catheters has been shown to delay bacterial colonization and thus reduce the incidence of bacteriuria. However, an RCT evaluating the use of two antibiotic-coated catheters did not find significant benefit in prevention of CAUTI. These catheters are more expensive and thus are likely not beneficial in most clinical settings. Use of a computerized clinical decision support intervention for reducing the duration of catheter use has been shown to be effective in limiting the duration of catheters and CAUTIs in hospitalized patients, although this intervention may be more difficult to implement in LTCFs.

As CAUTIs are the most common health care–associated infection, there have been regulatory changes directed toward incentivizing CAUTI reduction in both hospitals and nursing homes. PALTC settings are paying special attention to incorporating strategies for CAUTI prevention including making sure that residents have an indwelling catheter only when indicated. In one of the observational cohort studies including 228 long-stay residents with an indwelling catheter, 86% had a documented indication for the catheter. Of those 193 residents with a documented indication, 99% had one or more indications deemed appropriate, including urinary retention (83%), pressure ulcer (21%), hospice care (10%), and need for accurate measurement of input and output (6%). Evidence suggest that implementing intervention bundles that incorporate technical strategies (focusing on aseptic insertion, catheter maintenance, regular assessments, prompt removal when no longer indicated, and incontinence care planning) and socioadaptive strategies (focusing on effective communication and engaging nursing home leadership, staff, residents and their families in CAUTI prevention efforts) can result in significant reductions in CAUTI especially in those facilities struggling with higher CAUTI rates. When such intervention was implemented in 404 community nursing homes, their adjusted CAUTI rates decreased from 6.42 to 3.33 infections per 1000 catheter days (incidence rate ratio, 0.46; 95% CI, 0.36–0.58; $p < .001$). However, a similar intervention in Veterans Affairs nursing homes did not result in significant change (2.26–3.19 infections per 1000 catheter days; incidence rate ratio, 0.99; 95% CI, 0.67–1.44). The lack of success in further reduction of CAUTI rate was thought to be secondary to lower baseline CAUTI rate in this setting. Therefore, it is important for the nursing home leaders to evaluate their baseline CAUTI rates before dedicating resources specifically to a CAUTI-reduction bundle. Another study evaluated the impact of a multimodal targeted infection prevention program implementation in 12 community nursing homes. The program included active surveillance for infections, ongoing educational programs for staff, hand hygiene promotion, and preemptive barrier precautions for all residents with indwelling devices. This intervention resulted in both decreased MDRO prevalence density and CAUTIs. It provides evidence that CAUTI reduction effort can also be incorporated into more comprehensive infection prevention and control bundle in PALTC settings.

SPECIAL ISSUES

Outcomes

Most healthy older adults with symptomatic UTI respond well to antibiotic therapy. However, bacteriuria is associated with significant morbidity, often related to antibiotic use,

which promotes the development of multidrug-resistant bacteria and overgrowth of organisms such as *C difficile*. Presence of an indwelling catheter, age, and medical comorbidities influence outcomes. CAUTI was associated with a significantly higher mortality in hospitalized adults aged 65 or older compared to UTI in noncatheterized patients (19% vs 11%, $p < 0.05$). In a retrospective cohort study of residents with UTI in LTCFs in the United States, risk factors that appeared to increase the risk of hospitalization included advancing age, Hispanic or African-American race/ethnicity, diabetes mellitus, Parkinson disease, dementia, and stroke. Factors that decreased the risk of hospitalization in this same cohort included the ability to walk, physician visit at the time of admission to a LTCF, and maintaining or improving mobility in those with severe mobility problems. One of the most serious complications of UTI is the development of bacteremia as a consequence of persistent infection causing hematologic spread. Older adults are more likely than younger adults to develop gram-negative bacteremia as a consequence of UTI. The mortality in patients with bacteremic UTI has been described to be between 14% and 33%. However, adults with bacteremia as a result of UTI compared to bacteremia from other etiologies tend to have better outcomes.

Antibiotic Resistance

Antibiotic resistance is a large problem in all health care settings including outpatient, PALTC and acute hospital settings. UTI remains the most common reason antibiotics are prescribed in older adults, despite increased awareness of emerging resistance as a result of overtreatment. A substantial proportion of antibiotics are prescribed for asymptomatic infection or presentations without localizing urinary symptoms, fostering the development of increasingly resistant organisms.

Antibiotic resistance in the community Antibiotic resistance is no longer limited to the hospital setting. Primary care providers are responsible for a significant number of antimicrobial prescriptions in community-dwelling older adults and recent antibiotic use in the primary care setting has been shown to be the most important risk factor for development of an infection with a resistant organism. Adults who receive an antibiotic for UTI are over four times more likely to develop a resistant organism within 1 month compared to adults not prescribed an antimicrobial agent. Although the risk is greatest at 1 month, resistance persists for up to 12 months in many patients. A population-based cohort study in Minnesota found the incidence of resistant *E coli* urinary isolates to FQ plus TMP/SMX in adults aged 80 or older to nearly double over a 4-year period (274–512 per 100,000 person-years, $p < 0.05$). Despite the increasing awareness of drug-resistant pathogens in the community, overuse with antimicrobials for treatment of UTI remains. Community-dwelling older adults may believe that antibiotic resistance is a hospital-acquired condition and may not fully understand the risks associated with each antibiotic prescription. Primary care providers may feel that increasing resistance to antibiotics is largely out of their control. Further studies assessing patient and community physician understanding of antibiotic resistance, particularly as it relates to UTI, are needed.

Antibiotic resistance in PALTC settings Over the past several years, increased awareness has been placed on infection control practices and ASPs in LTCFs, although inappropriate use of antibiotics remains high in this setting. Prevalence rates of colonization with antibiotic-resistant bacteria have been reported to be higher than 50% in some LTCFs. Among all urinary isolates of residents in five nursing homes in Connecticut, 26% were resistant to TMP/SMX, 40% were resistant to FQs, and 25% were resistant to cefazolin. The resistance rates to FQ in only *E coli* isolates were much higher at nearly 60%. *E coli* isolates were highly sensitive to nitrofurantoin (93%), although its sensitivity to all organisms was much lower at 68%. Similar to the community setting, high antibiotic prescriptions driving increasing resistance rates continue to be prevalent despite increased awareness. Qualitative studies have characterized some variables, which drive inappropriate antimicrobial use in this setting. When antimicrobial therapy is prescribed, often, nursing personnel are evaluating patients and communicating their findings to physicians. Physicians may not assess patients directly. Nursing personnel (ie, nurses and nurses' aides) are critical to identifying patients that are suspected of having UTI, requesting urine studies to be sent, and suggesting that antibiotic therapy may be warranted. Interventions focused on educating nursing staff to perform detailed assessment of a resident when a UTI is suspected and improving communication with physicians after assessment have resulted in decreased antibiotic use for the UTI in PALTC settings.

Antimicrobial Stewardship Programs

UTIs are major driver of antibiotic use and inappropriate antibiotic prescribing related to UTI is common in various health care (including acute-care, outpatient, and PALTC) settings. ASPs are designed to promote appropriate antibiotic prescribing while reducing the adverse events associated with antibiotic use. CDC has published core elements of antibiotic stewardship programs for different health care settings. For hospitals and nursing homes, there are seven recommended core elements including (1) leadership commitment, (2) accountability, (3) pharmacy expertise (for hospitals)/drug expertise (for nursing homes), (4) action, (5) tracking, (6) reporting, and (7) education. For outpatient antibiotic stewardship, they outlined four core elements including (1) commitment, (2) action for policy and practice, (3) tracking and reporting, and (4) education and expertise. CDC's core elements of antibiotic stewardship provide a framework that clinicians and health care facilities can use to put in place comprehensive set of interventions that involves reviewing antimicrobial stewardship related data (eg, antibiotic use and outcome data),

identifying opportunities for improvement, implementing action plan for improvement in antibiotic prescribing practices, and providing periodic updates and feedback to health care workers on the progress. If the initial evaluation of the data points toward UTI as the major driver of the inappropriate antibiotic prescribing in the facility, then ASP should consider focusing its attention on interventions that can promote appropriate antibiotic use for UTIs. The exact intervention needed may depend on underlying reason for inappropriate prescribing. For example, if antibiotics are being initiated for UTI in scenarios where the patients are not meeting the clinical criteria for the diagnosis of UTI, then introducing an assessment or decision-aid tool prompting clinicians to look for those criteria during initial evaluation will be a good intervention. If the reason for inappropriateness is unnecessary use of broad-spectrum antibiotics or use for longer than recommended duration, then introducing facility-specific guidance for clinicians outlining the preferred empiric regimen and duration will be helpful. **Figure 106-3** highlights some examples of UTI-focused interventions that can be implemented in various health care settings. Depending on the identified challenges, a combination of interventions might be necessary for reducing inappropriate antibiotic prescribing and improving patient outcomes. One of the recent studies conducted in 25 nursing homes (12 intervention and 13 control) implemented a multifaceted antimicrobial stewardship program focused on reducing unnecessary antibiotic use for uncomplicated cystitis. Intervention nursing homes were randomized to receive a 1-hour introductory webinar, posters and pocket-sized educational cards with the diagnostic and treatment guidelines, tools for system change (eg, standardized physician order set forms; and an active monitoring sheet), and educational clinical vignettes addressing the diagnosis and treatment of suspected uncomplicated cystitis. Monthly web-based coaching calls were held for staff of intervention nursing homes and the facilities received quarterly feedback reports regarding management of uncomplicated cystitis. Overall antibiotic use was lower and fewer unlikely cystitis cases were treated in the intervention group without any increase in all-cause hospitalizations or death. Furthermore *C difficile* infection rates were also lower in the intervention as compared to the control nursing homes.

Evaluation and Management of UTI Near End of Life

Antimicrobials are usually overused at end-of-life and in the setting of advanced dementia, often without a known source of infection. That being said, UTIs can carry a heavy symptom burden. Thus, the decision to evaluate symptoms and subsequent management of a confirmed UTI are influenced greatly by the goals of care and life expectancy of the individual. If a patient is able to reliably report symptoms, and is expected to survive long enough to benefit from antimicrobial treatment, it would be appropriate to consider routine care. Enrollment in hospice alone should not be a barrier to receiving work-up and treatment of UTIs. However, antibiotic therapy has risks along with the potential benefits for a patient and should not be presented as a benign treatment plan. A discussion of potential side effects, risk for infection with resistant organisms, and any medical monitoring that is required for the selected antibiotic

Outpatient	Hospital	Long-term care facility
• Implement strategies (eg, pocket cards, educational posters, etc) to promote use of established diagnostic criteria and evidence-based treatment recommendations • Communications skills training for clinicians to better prepare them for discussions with patients and families on appropriate antibiotic use • Incorporate evidence-based guidelines into electronic health record order sets • Establish urine culture follow-up program • Provider feedback reports	• Facility-specific diagnosis and/or treatment guidelines • Prospective audit and feedback • Requiring prior authorization for certain class of antibiotics (eg, fluoroquinolones) • Antibiotic time-out • Implementing intravenous to oral conversion protocols • Pharmacy-driven dose adjustment and optimization	• Implement SBAR (situation – background – assessment – recommendation) tool for effective communication and management • Establish standing orders for active monitoring for nonspecific signs and symptoms • Facility-specific diagnosis and/or treatment guidelines • Implement mandatory review of necessity by medical directors for all outside antibiotic orders • Antibiotic time-out • Provider feedback reports

FIGURE 106-3. Examples of interventions that can be implemented by healthcare settings as part of antibiotic stewardship program.

should be presented so patients can make a well-informed decision.

Advanced dementia, aphasia, or any disease process which interferes with a patient's ability to provide a reliable history can confound efforts to assess symptoms of infection, such as dysuria. If a UTI is suspected, it is important for clinicians to have a good understanding of the patient's wishes for their care prior to further investigation. Patients or their surrogate decision-makers may have set limits on what treatments they would want in such circumstances. If the goal is comfort, then it may be most appropriate to defer a work-up and focus on palliation of symptoms alone. If no limits have been set, or there is confusion about what a patient may or may not have wanted in this clinical setting, factors such as evidence of the seriousness of the infection, overall life expectancy and likelihood of survival benefit from antimicrobial treatment should be discussed with the medical decision-makers. When deciding against additional work-up and antibiotic treatment for a suspected UTI, focus can be directed toward symptom management strategies.

CONCLUSION

UTI is the most common infection occurring in older populations. The management of UTI for older ambulatory individuals is similar to younger populations, and determined by the presentation as either acute simple cystitis, complicated UTI or CAUTI. Treatment of symptomatic UTI requires optimal use of urine culture for diagnosis, appropriate antimicrobial selection, and appropriate duration of therapy. In PALTC settings, the very high prevalence of ASB interferes with ascertainment of symptomatic infection, especially in residents without localizing genitourinary symptoms. Bacteriuric older individuals who present with clinical decline without localizing findings should not be assumed to have a urinary source, despite a positive urine culture. A critical approach to diagnosis, investigation, and treatment is necessary. Further characterization of symptomatic presentations of urinary infection in the functionally disabled older population, together with evaluation of management approaches such as nontreatment with observation compared with empiric antimicrobial treatment, is needed.

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Chapter 107

Other Viruses: Human Immunodeficiency Virus Infection and Herpes Zoster

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HUMAN IMMUNODEFICIENCY VIRUS INFECTION

Learning Objectives

- Recognize the shift in the epidemiology of human immunodeficiency virus (HIV) over the last three decades.
- Identify older adults at risk for HIV infection and include HIV testing in routine care.
- Recognize that some comorbidities may present earlier or have a higher prevalence among older adults with HIV.
- Integrate the principles of geriatrics (eg, management of multimorbidity and polypharmacy) into the care of older adults with HIV.

Key Clinical Points

1. **Antiretroviral therapy (ART) has turned HIV into a chronic disease that one ages with over many decades. More than 50% of all people living with HIV in the United States are age 50 or older.**
2. **HIV is often unrecognized in older adults because of subtle or nonspecific presentation (anemia, fatigue, weight loss, pneumonia) and provider bias that older adults are not at risk.**
3. **While AIDS-associated conditions have declined markedly, comorbid illnesses occur more frequently than in age-matched, HIV-uninfected control populations leading to early multimorbidity, polypharmacy, and functional decline even in those with HIV controlled by ART.**

DEFINITION

HIV is the virus that causes acquired immunodeficiency syndrome (AIDS). HIV destroys CD4+ T cells vital to defense of infections, and when untreated, leads to a progressively immunocompromised state with an increased risk for opportunistic infections, cancers, and other complications.

EPIDEMIOLOGY

One of the most amazing success stories of our lifetime is the transformation of HIV infection from a rapidly fatal disease that manifested in the first AIDS cases in early 1980s to a chronic health condition, with this shift driven in part by the tremendous advocacy of the HIV community. With early initiation of effective ART, HIV-1 RNA can now be quickly reduced to undetectable levels. Not only does this decrease the risk of immune suppression and associated comorbidities, but effective ART essentially eradicates between person transmission of HIV. This discovery that Undetectable = Untransmittable (or U = U) has resulted in a transformative experience for many people living with HIV, as they can now live without the fear of transmitting HIV to partners or close contacts. Unfortunately, despite early initiation of ART, we have yet to fully eradicate HIV, with rare exceptions occurring in the setting of bone marrow transplantation. As a result, a growing number of people are now living, and aging, with HIV. In the United States and Europe, as of 2018, over 50% of people living with HIV are age 50 or older, with approximately 20% age 65 or older.

In addition to aging with HIV, many older adults continue to become newly infected with HIV: more than 15% of new HIV diagnoses in the United States are in adults aged 50 or older. Those aging with HIV are more likely than other aging individuals to use psychoactive substances including alcohol, marijuana, and opioids, which can both interfere with antiretroviral adherence and increase higher-risk

sexual encounters. Use of medications to treat erectile dysfunction and enhance sexual performance may also increase sexual risk behavior. Additionally, older men are less likely to use condoms and postmenopausal women may be less likely to request barrier protection without the risk of pregnancy. Age-associated erectile changes may make condom use difficult, and age-associated declines in immunity and changes in mucosal barriers may place older individuals at higher risk of transmission with each exposure. Finally, older individuals often present later in the course of HIV disease, extending the period of highest infectivity due to unsuppressed HIV-1 RNA.

While uptake of pre-exposure prophylaxis (PrEP) is less common among older versus younger adults, PrEP is an extremely effective tool for prevention of HIV transmission among patients at risk of HIV acquisition. Efficacy is presumed to be similar among older adults; however, few studies have included older adults, and the risk of adverse events and toxicity may be higher. A combination of strategies (barrier protection, ensuring viral suppression among people with diagnosed HIV, and providing PrEP to people at risk for HIV) has decreased and can continue to decrease transmission of new cases of HIV.

EVALUATION

Screening

No one is immune to HIV infection, and nearly 15% of people with HIV (PWH) in the United States and approximately 45% of persons worldwide are unaware of their diagnosis, presenting the greatest transmission risk. The nonspecific or asymptomatic presentation of new HIV may lead to a missed diagnosis for months or even years, as physicians and their patients are both likely to underestimate the risk of HIV infection among older adults. Prompt diagnosis and treatment of HIV decreases transmission and improves outcomes across the age spectrum. U = U has also led to a greater emphasis on the HIV care continuum: ensuring that each person living with HIV is diagnosed, linked to and receives care, is retained in care overtime, and achieves viral suppression.

The Centers for Disease Control and Prevention and the US Preventive Services Task Force (USPSTF) currently recommend that adolescents and adults aged 13 to 64 (CDC) or 15 to 65 (USPSTF) and all pregnant women should have HIV screening at least once; other expert panels (including the American College of Physicians) recommend screening up to age 75. More frequent annual testing is recommended for adults with higher risk, including persons who use injection drugs; have sex with a partner who has HIV or unknown HIV status, or uses injection drugs; and men who have sex with men. An individual in a monogamous relationship may present with a sexually transmitted disease because their partner has more than one partner; any patient with a sexually transmitted disease (gonorrhea, syphilis, chlamydia, or trichomonas) should be screened (or rescreened) for HIV. Older patients may not spontaneously report sexual or drug use behaviors, and physicians seldom ask. Notably, some older patients may assume that HIV testing is part of routine blood work, thus incorrectly believe they are regularly tested. Annual wellness visits or hospitalizations are opportunities to ensure that HIV testing is up-to-date and repeated if indicated. Specific signed consent is no longer required for HIV testing though verbal consent may be required in some states.

In addition to routine testing, HIV testing should be conducted as part of a diagnostic work-up in any patient with unexplained laboratory results (anemia, leukopenia, pancytopenia, transaminitis, hyperproteinemia); new diagnoses of peripheral neuropathy, oral candidiasis, herpes zoster, recurrent bacterial pneumonia, anal cancer; or any AIDS-associated conditions (*Pneumocystis jiroveci* pneumonia, Kaposi sarcoma, atypical mycobacterial infection, tuberculosis, non-Hodgkin lymphoma). Finally, any patient presenting with debilitating fatigue, weight loss, or cognitive changes should be tested for HIV.

HIV Testing

In most clinical settings, the prior two-step ELISA (antibody) and Western blot testing strategy has been replaced by a fourth-generation combination HIV-1/2 immunoassay that tests for both HIV-1 and HIV-2 antibodies and the HIV p24 antigen. If positive, a confirmatory immunoassay will distinguish between HIV-1 and HIV-2 (primarily found in Africa). If there is any concern for recently acquired HIV infection, based on recent exposure or symptoms (fever, malaise, lymphadenopathy), the immunoassay should be combined with a PCR test for HIV-1 viral load as the p24 antigen will not turn positive until approximately 5 to 7 days after the HIV-1 viral load is detectable. Both false-negative and false-positive tests are now extremely rare with the combined antibody/antigen test. Rapid screening tests (including home kits) detect antibody only and have a higher rate of false negative results.

MANAGEMENT

Selection and Initiation of Antiretroviral Treatment

Patients with a new HIV diagnosis should be connected to HIV care as quickly as possible. ART is recommended for any patient regardless of CD4+ T-cell count, with greatest benefit on long-term prognosis seen if the CD4+ count remains more than 500 cells/μL. Rapid treatment of HIV offers immediate benefit by preventing declines in CD4+ count and decreasing HIV transmission. Indeed, some HIV programs have instituted rapid start programs with initiation of ART on the same day of HIV testing. Data also

suggest that rapid initiation of ART with lower peak HIV-1 viral loads may decrease the "reservoir" of HIV that is difficult to eradicate and may have added benefits on long-term inflammation and immune alterations with aging. Older adults appear to adhere more effectively to ART and are quite successful at achieving HIV-1 RNA suppression, but experience a slower improvement in CD4+ cell count, thus even more emphasis is placed on early initiation. Without treatment, the natural history of HIV infection is usually one of progressive immunodeficiency, with development of symptomatic AIDS approximately 8 to 12 years after infection. Prognosis is poorer among those who present late for care, fail to adhere to their treatment regimen, and/or develop widely resistant virus.

Standard ART regimens require medications from at least two different classes, most commonly an integrase strand transferase inhibitor (INSTI) combined with one or two nucleoside reverse transcriptase inhibitors (NRTIs). Compared to older antiretrovirals, contemporary regimens have less adverse effects, fewer drug–drug interactions, rapid viral suppression, and are generally tolerant of minor variations in adherence. Furthermore, many of the regimens are available in a single coformulated tablet once a day. Long-acting injectable drugs also have promising data on adherence and acceptability, though with very limited data among older populations. Notably, while ART appears safe in older populations and is essential in the prevention of disease progression, very few adults older than age 60 were included in clinical trials.

There are many antiretrovirals from which to choose, and this selection should be informed by consultation with an experienced HIV care provider. Selection of a regimen incorporates likelihood of resistance (and HIV-1 genotyping if available), cost, patient preference, comorbidities, preexisting organ system injury and anticipated toxicity, and concomitant medications. Patients are followed closely for the first several months of therapy to ensure tolerability, safety, and HIV-1 viral load suppression. After the first 6 to 12 months of therapy, if viral suppression is achieved and the patient has a CD4+ count above 350 to 500 cells/mL, HIV-1 viral load monitoring can occur approximately every 6 months and CD4+ count annually or less. While selection of antiretroviral regimens should be informed by an experienced HIV provider, once patients have achieved HIV-1 viral load suppression, care could be managed by a primary care provider or geriatrician with consultation as needed.

Many toxicities with ART are predictable. Weight gain is common with any regimen, but appears to be more common and significant with newer antiretroviral regimens. While some weight gain may be beneficial among those who are underweight, in many, weight gain is associated with incident hypertension, hyperlipidemia, diabetes, and hepatic steatosis. Thus, all patients initiating ART should be provided general nutrition and physical activity recommendations to avoid excess weight gain. Physical activity prescriptions specific for adults with HIV have been developed and can be given at the time of ART to provide strategies to maintain adequate levels of physical activity.

General Care Considerations for the Older Adult With HIV

After antiretroviral initiation, the incidence of AIDS conditions drops dramatically, but ongoing and consistent therapy is imperative. The Strategies for Management of Antiretroviral Therapy (SMART) trial demonstrated that continued viral suppression decreases incident non-AIDS conditions including liver disease, kidney disease, lung disease, and cardiovascular disease. Even with early initiation of ART, PWH experience some comorbidities many years before their uninfected peers (accelerated), or with a much higher prevalence than their uninfected peers (accentuated). This concept of accelerated versus accentuated aging is particularly important for diseases that might occur earlier and require a change in screening recommendations (eg, low bone density, falls, frailty, some cancers), and for diseases that occur more frequently and may require particular vigilance for monitoring and aggressive management (eg, cardiovascular disease and certain cancers). With the earlier manifestation of many diseases, a threshold of 50 years has been used for characterizing "aging" with HIV. As the population of people older than 60 or 65 years continues to grow, it will be important to also study effects of HIV among those of more advanced age.

Potential mechanisms underlying accelerated or accentuated aging in PWH include immune dysfunction, dysregulation, and suppression; chronic inflammatory stimulation due to reservoirs of HIV-1 RNA replication and chronic microbial translocation; hypercoagulability; and direct effects of viral proteins, modifiable lifestyle factors, and social or structural determinants of health. Many of the immune system changes of senescence in aging mirror the changes seen with HIV. Furthermore, markers of aging such as DNA methylation patterns and shortened telomere length indicate that PWH are approximately 5 years older, biologically, than their HIV-uninfected peers. Those who acquire HIV infection often have had years of struggle with structural inequalities (homelessness; biases due to race, ethnic, and gender minority populations; employment; insurance), contributing to additional stress-related immune responses.

While these mechanistic changes may occur to some extent among almost all PWH, the disease process, immune system alterations, and comorbidities of someone who has lived with HIV for over 30 years may be much different than someone diagnosed more recently and started on therapy immediately. The "legacy" effect of immune suppression, opportunistic infections, and exposure to older, toxic ART may be long-lasting. In contrast, those

that survived the early years of HIV, while experiencing the stress and trauma of the AIDS epidemic, may also have a unique "survivor" effect. Individuals who are diagnosed early in their disease process, achieve and maintain excellent HIV-1 RNA suppression, do not gain excess weight after ART initiation, and avoid ongoing substance use are likely to experience a very different aging trajectory from those who do not.

Models of Geriatric HIV Care

Accompanying an increasing number of older adults with HIV is, unfortunately, a shortage of both HIV and geriatric medicine clinic providers. In addition to routine clinic-based care, people aging with HIV are substantially more likely to be hospitalized than their HIV-uninfected peers. How these patients and their many transitions between inpatient and outpatient care, general medical care and specialty care, and skilled nursing facility care will best be managed in the future remains to be seen. Incorporation of telehealth or telephone consultation for geriatric, HIV, or other subspecialty care may be an important component of care in the future. Similarly, HIV providers should have appropriate training in geriatric medicine, and geriatric care providers in HIV medicine. What is nearly certain is that older PWH will place new demands on parts of the health care system, including short- and long-term nursing facilities that have previously not managed those with HIV infection in any substantial numbers.

While the management of most diseases will be the same among older PWH, providers should be cognizant of diseases that occur more frequently or manifest differently in HIV. As in other geriatric populations, consideration of the 5 M's approach to care can prioritize key clinical issues among many PWH, with a focus on Mind, Multimorbidity, Medications, Mobility, and (what) Matters most. We have proposed adding a sixth M for modifiable, emphasizing the importance of addressing the still *modifiable* factors in a population that might be younger than the typical geriatric population.

Mind incorporates cognitive impairment, mood disorders, the impact of substance use, and the added impact of stressors such as HIV stigma, loneliness, and depression. PWH have almost a fivefold risk for developing cognitive impairment independent of age, and may experience both HIV-related impairments and age-associated cognitive disorders such as Alzheimer disease, mild cognitive impairment, and vascular dementia. Up to 50% of PWH (of any age) will have evidence of HIV-associated neurocognitive impairment on neuropsychiatric testing, and approximately one-quarter experience an impact on daily function. Importantly, the Mini-Mental State Examination may miss the executive function impairment most common in HIV-associated neurocognitive disorders; the Montreal Cognitive Assessment is a more sensitive screening tool, though formal neuropsychiatric testing is generally recommended. Although HIV disease severity (low CD4+ count, high HIV-1 RNA) is strongly linked to HIV-associated neurocognitive disorder, the impairments have changed minimally despite early and consistent ART; comorbidities, medication toxicity, psychiatric disease, and substance use also play an important role. Depression, anxiety, and severe mental illnesses such as psychosis and posttraumatic stress disorder are more common among older PWH infection than among the general population, with depression occurring in over 50% of PWH in some cohorts. Mood disorders can overlap with and confound the diagnosis of cognitive impairment, thus teasing out the diagnosis of both can be challenging. PWH who are aged 50 or older are also more likely than uninfected demographic counterparts to consume hazardous amounts of alcohol, use illicit drugs (especially marijuana, cocaine, and opioids), and smoke cigarettes. Lastly, social isolation, loneliness, and HIV-related stigma are highly prevalent among older PWH and may contribute to both poor cognition and mental health.

PWH experience a greater burden of *multimorbidity* at a given age than other populations, particularly in regard to clusters of cardiovascular and metabolic diseases, including diabetes, hypertension, hyperlipidemia, and nonalcoholic fatty liver disease. These disease clusters may be linked to toxicities of older ART, weight gain associated with newer therapy, and chronic inflammation. Kidney disease and loss of bone density are particularly common with some ART regimens and may occur as a consequence of coinfection with hepatitis B or C. The risk of some cancers is heightened due to coinfection with oncogenic viruses (human papilloma virus and anal cancer), exposure to carcinogens (lung cancer and smoking), or immunosuppression (some lymphomas). When referring to specialists, it is important to recognize that many older PWH continue to experience stigma related to HIV, gender, sexual preference, and age, and may be resistant to evaluation in a new clinic with a new provider.

With the initiation of ART, *multiple medications* or polypharmacy becomes the "norm" among PWH, often at least a decade before those without HIV experience polypharmacy. In addition to the burden of daily medications and pill fatigue, many antiretroviral agents are associated with significant drug–drug interactions. The protease inhibitors are particularly problematic, increasing levels of some concomitant medications (such atorvastatin) by over 800%. With the introduction of newer ART, treatment guidelines have emphasized a shift away from certain medication classes, particularly among older PWH, to avoid these often dangerous drug interactions. Interactions with food, vitamins, and both prescription and nonprescription medications can still occur with almost any of the HIV medications and careful attention to possible drug interactions should be considered with any new

prescription or change in dose. Further, due to preexisting organ system injury among those with HIV, those with HIV infection may be particularly susceptible to the harms associated with polypharmacy. These harms include decreased medication adherence and increased serious adverse drug events including organ system injury, hospitalization, geriatric syndromes (falls, fractures, and cognitive decline), and all-cause mortality. Utilizing pharmacists and drug–drug interactions programs (eg, HIV-druginteractions.org) can be extremely helpful.

Mobility tends to decline at a faster rate among PWH compared to uninfected controls, with differences apparent as early as age 50. Even among those with early initiation of ART, impairments in measures of grip strength, lower extremity function (repeat chair stands), and balance are common. The effects of ART on mitochondria, inflammation, neuropathy, obesity, and sarcopenia are heightened further by the impact of comorbidities, socioeconomic factors, and physical inactivity. Importantly, mobility impairments in middle-aged and older PWH are linked strongly to poor outcomes, including increased morbidity, mortality, and risk of falls. Indeed, people aged 45 to 65 with HIV experience fall rates similar to populations aged 65 and older without HIV. Many of the mobility impairments can be attenuated with cardiovascular and strength training. Thus physical activity counseling and fall risk reduction should be an important component of the care of PWH, regardless of age.

Ultimately, care should address what *Matters Most* to the patient. Recommendations should be informed by prognosis and the added risk/benefit ratio. Greater comorbid burden, frailty, and cognitive impairment among a relatively younger PWH may limit the added benefit of a screening procedure or intervention; other older adults with HIV may have robust health with few comorbidities. For example, patients need to live at least 5 to 7 years after screening for the benefit of the screening (decreased mortality) from colon cancer to outweigh the risk of perforation and the pain and discomfort of the procedure. Conversely, more aggressive screening for anal cancer may be indicated, particularly among those with good health, because of the increased incidence of this condition among those with HIV infection. Thus, "age appropriate" primary care guidelines for those without HIV may differ among PWH—careful consideration for the individual's prognosis, preferences for care, and personal risk of these conditions should be incorporated in the conversation. What matters most in the care of an older PWH may be impacted considerably by loss, trauma, and stigma. Importantly, these conversations and designated decision makers should be carefully documented. Less than half of patients in most HIV clinics have completed any form of advanced care planning. Many PWH in the United States have estranged relationships with family and have same sex partners or close friends that they would prefer to designate as a medical power of attorney. As the recognition of these decision makers can differ by state, addressing advanced directives and designating medical power of attorney among older PWH is even more important to ensure the right person can help to make the right choices at the end of life.

Lastly, *modifiable* risk factors should be a key component of the care of PWH throughout the lifespan. Some older PWH may continue to engage high-risk sexual behaviors and have frequent exposures to sexually transmitted infections (STIs); safer sex and decreasing long-acting consequences of STIs, including syphilis, hepatitis B, and hepatitis C. Food insecurity is up to six times more common in older adults with HIV, and loneliness occurs in nearly 60% of older adults with HIV: ensuring access to nutritious food and connecting with social resources may improve multiple domains of health. As previously mentioned, consistent ART adherence and adequate levels of physical activity remain key components of care for all PWH.

In summary, people aged 50 or older now account for a majority of PWH as a consequence of improved lifespan of those with HIV and increased acquisition of HIV at older ages. Older adults should be screened for HIV at least once, and more often if sexual risk is unknown or the patient has nonspecific symptoms. As a consequence of biobehavioral factors, HIV or ART-associated factors, structural inequalities, and the impact of aging-related factors, PWH experience a greater burden of comorbidities and develop geriatric syndromes, often at a much earlier age than their HIV-uninfected peers. A 6 M's approach to the older PWH may guide providers in prioritizing care components (**Figure 107-1**). Importantly, the study of HIV infection among aging individuals may provide a template for improving the management of complex chronic disease more generally, especially among special populations of aging individuals—people of color, sexual minorities, and those with limited socioeconomic resources.

Biobehavioral factors among PWH
- Physical inactivity
- Poor nutrition
- Obesity
- Smoking or other substance use
- Risky sexual behaviors
- HCV exposure/coinfections

HIV- or ART-associated factors
- Inflammation
- Immunosenescence
- Epigenetic alterations
- Mitochondrial toxicity
- Microbial translocation
- Adipose tissue dysfunction

Pillars of aging without HIV
- Immunosenescence & stem-cell exhaustion
- Telomere shortening
- Epigenetic alterations
- Dysregulated nutrient sensing
- Mitochondrial dysfunction
- Loss of proteostasis
- Altered intercellular communication

Comorbidities in older PWH
- Cardiometabolic
- Cancer
- Osteoporosis
- Renal and liver disease
- Mood disorders

Structural inequalities among PWH
- Race/ethnicity
- Gender
- Education/housing
- Employment/access to insurance
- Stigma
- Stress/trauma

Geriatric syndromes
- Polypharmacy
- Neurocognitive impairment
- Frailty
- Sarcopenia
- Falls

6 M's approach to prioritizing care of older PWH
- Mind (cognition, mood)
- Multimorbidity (disease clusters)
- Medications (polypharmacy or inappropriate therapy for older adults)
- Mobility (gait speed, strength, balance, falls)
- Matters most (goals of care, advanced directives)
- Modifiable (physical activity, nutrition, access to services)

FIGURE 107-1. Underlying factors influencing development of comorbidities and geriatric syndromes and considerations to the approach of care of the older adult with HIV. Numerous factors contribute to the development of comorbidities and geriatric syndromes among older adults with HIV (blue boxes) including the pillars of aging, behavioral factors, HIV, or antiretroviral therapy (ART) associated, and structural inequalities that may be more common among people with HIV (PWH). Many comorbidities (green box) and geriatric syndromes (orange box) appear to occur earlier or more common among PWH. These contributing factors, comorbidities, and geriatric syndromes may be best approached through a 6 M's approach to care, as summarized in the purple box. HCV, hepatitis C virus.

HERPES ZOSTER

Learning Objectives

- Understand the impact of aging on the incidence of herpes zoster.
- Recognize typical and atypical presentations of herpes zoster in older adults.
- Know when to order microbiological testing for the diagnosis of herpes zoster and what tests to order.
- Employ the appropriate use of different treatments for the management of herpes zoster.
- Summarize Advisory Committee on Immunization Practices (ACIP), Centers for Disease Control and Prevention (CDC), recommendations for the use of the recombinant zoster vaccine.

Key Clinical Points

1. **All cases of herpes zoster are caused by the reactivation of endogenous varicella-zoster virus (VZV) from latent infection of dorsal sensory or cranial nerve ganglia and not by exogenous transmission of VZV from an individual with herpes zoster or varicella.**
2. **Increasing age and impaired cellular immunity are the strongest risk factors for herpes zoster.**
3. **Although the diagnosis can be made on clinical grounds in the majority of cases with a typical dermatomal vesicular rash and pain, laboratory diagnostic testing is indicated when differentiating herpes zoster from**

(*Continued*)

(Cont.)

HSV, for suspected organ involvement, and for atypical presentations, particularly in the immunocompromised host; polymerase chain reaction (PCR) of vesicular fluid is the preferred diagnostic test.

4. The goal of the treatment of herpes zoster in older adults is to decrease the length of the acute attack and to reduce pain by the use of early antiviral therapy (acyclovir, famciclovir, valacyclovir), scheduled analgesia, and if the pain is not adequately controlled, adjunctive agents.
5. The recombinant zoster vaccine is recommended for all immunocompetent adults 50 years and older by ACIP for the prevention of herpes zoster.

DEFINITION

Herpes zoster is a neurocutaneous disease that is caused by the reactivation of VZV from a latent infection of dorsal sensory or cranial nerve ganglia following primary infection with VZV earlier in life. Herpes zoster is characterized by unilateral, dermatomal pain and rash. Postherpetic neuralgia (PHN) is pain 90 or more days after rash onset.

EPIDEMIOLOGY

VZV Transmission

VZV is a double-stranded DNA herpesvirus that is transmitted from person to person via direct contact or airborne or droplet nuclei when a virus-naive individual is exposed to the vesicular rash of varicella or herpes zoster. These exposed individuals may then develop varicella. Health care workers and staff in nursing homes and hospitals and children who have not received the varicella vaccine may not have had VZV primary infection and are potentially at risk for varicella. However, nearly all older adults are seropositive and latently infected with VZV. The exposure of a latently infected individual to herpes zoster does not cause herpes zoster or varicella. All cases of herpes zoster are caused by the reactivation of endogenous VZV.

Herpes Zoster Incidence and Risk Factors

In general population studies from North America and Europe, the estimated incidence of herpes zoster in persons older than age 65 varies from approximately 10 to 14 cases per 1000 per year. The lifetime incidence of herpes zoster is estimated to be about 30% in the general population and may be as high as 50% among those surviving to 85 years or older. Current population figures and herpes zoster incidence data yield estimates of about 1 million new cases of herpes zoster each year in the United States.

The cardinal epidemiological feature of herpes zoster is its increase in incidence with aging and with diseases and drugs that impair cellular immunity. The increase in the likelihood of herpes zoster with aging starts around 50 to 60 years of age and increases into late life in individuals older than 80 years (**Figure 107-2**). Immunocompromised patients at risk for herpes zoster include persons with HIV infection, Hodgkin disease, non-Hodgkin lymphomas, leukemia, hematopoietic stem cell or solid organ transplant, systemic lupus erythematosus, and those taking immunosuppressive medications. Other risk factors

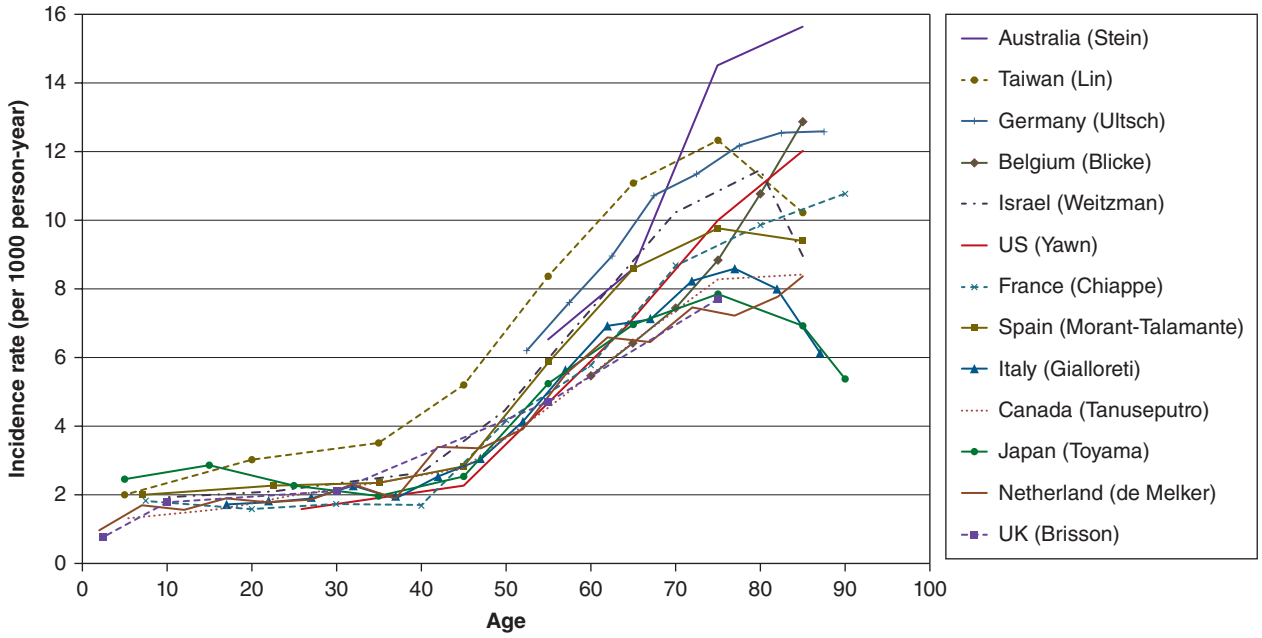

FIGURE 107-2. Age-specific incidence rate of herpes zoster in North America, Europe, and Asia-Pacific.

include White race, female sex, psychological stress, and physical trauma.

PRESENTATION

VZV reactivates in the affected sensory ganglion and spreads in the peripheral sensory nerve, causing prodromal symptoms that frequently precede the rash by several days. These prodromal symptoms are located in the involved dermatome and range from a superficial itching, tingling, or burning to severe, deep, boring, or lancinating pain, paresthesia, or hyperesthesia. The pain can be constant or intermittent. The prodrome may be mistaken for many other conditions in older adults, including pleurisy, a myocardial infarction, cholecystitis, appendicitis, renal colic, a collapsed intervertebral disk, glaucoma, trigeminal neuralgia, or unappreciated trauma. The prodromal symptoms usually last a few days, although they have been reported to last more than a week. Zoster sine herpete is a condition when patients experience acute dermatomal neuralgia without ever developing a skin eruption.

When VZV reaches the dermis and epidermis, the initial cutaneous presentation of herpes zoster occurs as a unilateral, dermatomal, red, maculopapular eruption. Vesicles generally form within 24 hours and evolve into pustules, which dry and crust in 7 to 10 days. The rash usually heals over in 2 to 4 weeks. In normal individuals, new lesions may continue to appear for 1 to 4 days and occasionally for as long as 7 days. The T1 to L2 or V1 dermatomes are most commonly involved (**Figure 107-3**). Atypical rashes may occur in older persons. The atypical rash may be limited to a small patch located within a dermatome or may remain maculopapular without ever developing vesicles.

Herpes zoster–associated acute neuritis produces dermatomal neuralgic pain in many older adults. Some older patients never develop pain, while others may experience the delayed onset of pain days or weeks after rash onset. The neuritis is described as burning, deep aching, tingling, itching, or stabbing. A subset of patients may develop severe pain, especially those with trigeminal nerve involvement. Acute herpes zoster pain can interfere with all activities of daily living (ADLs), but interference is greatest for instrumental ADLs, as well as sleep, general physical activity, and leisure activities. As overall pain burden increases, patients experience poorer physical, role, and emotional functioning.

FIGURE 107-3. A patient with herpes zoster illustrates a range of lesions (eg, vesicles, populates, and crusting).

The most feared complication of herpes zoster among older adults is PHN. PHN patients describe the quality of their pain as burning, throbbing, stabbing, shooting, sharp, aching, gnawing, tiring, or tender. The pain may be spontaneous and/or stimulus-evoked. Spontaneous pain may be constant aching or burning and/or brief, intermittent shock-like sensations. Stimulus-evoked pain consists of allodynia or hyperpathia. Allodynia is pain elicited by an innocuous stimulus and is particularly problematic. A cold wind, clothes, or bed sheets against the allodynic skin may cause debilitating pain. Hyperpathia is an exaggerated pain after a mildly painful stimulus. A minor bump of the affected dermatome against an object can cause severe pain that lasts for hours. Patients may experience discomfort that is not described as pain such as numbness or itching. Postherpetic itch can be maddening and appears to be mediated by damage and dysfunction of peripheral and central itch-specific neurons.

PHN greatly interferes with the functional status and quality of life of older adults. Patients can suffer from a variety of constitutional symptoms including chronic fatigue, anorexia, weight loss, and insomnia. PHN is associated with depression and social isolation in older adults. Reduced quality of life and interference with ADLs increase significantly as pain severity increases. Instrumental and basic ADLs are all compromised by PHN. For example, the patient with allodynic skin may be forced to avoid bathing or clothing around the affected area. Instrumental ADLs commonly affected include traveling, shopping, cooking, and housework. Studies using standardized pain scales and measures of function and quality of life such as the SF-36, SF-12, EuroQol, and Nottingham Health Profile have shown that PHN significantly interferes with multiple ADLs, reduces health-related quality of life (HRQL), and impairs mental and physical health.

Other complications include ocular inflammation in the setting of ophthalmic herpes zoster, cranial neuropathies, myelitis, visceral involvement, and motor paralysis. VZV-induced damage to the cornea and uvea and other eye structures can cause corneal anesthesia and ulceration, glaucoma, optic neuritis, eyelid scarring and retraction, visual impairment, and blindness in patients who did not receive antiviral therapy. Ophthalmic herpes zoster is also associated with stroke occurring weeks to months after the onset of the rash. Involvement of the third division of cranial nerve V can produce lesions and pain in the mandibular

area of the face and in the mouth. Cranial nerve VII and VIII involvement, or Ramsey Hunt syndrome, is characterized by facial palsy in combination with herpes zoster lesions of the ipsilateral external ear, ear canal, tympanic membrane, or hard palate. Symptoms include tinnitus, vertigo, deafness, problems with balance, and facial paresis. Involvement of cranial nerves IX and X may produce pharyngeal or laryngeal lesions and resulting pharyngitis or laryngitis. Motor paralysis, from VZV-induced destruction of motor neurons in the anterior spinal horn or motor radiculitis corresponding to the extension of VZV infection from the involved sensory ganglion, involves muscle groups with innervation that is contiguous with that of the affected dermatome. Painful Bell palsy in an older adult strongly suggests herpes zoster.

EVALUATION

In patients who present with prodromal pain and no rash, herpes zoster is often confused with other causes of unilateral, localized pain in older adults. One diagnostic clue to the presence of prodromal pain is localized cutaneous sensory abnormalities (eg, hyperesthesia, dysesthesia) in the affected dermatome. Herpes zoster is easily diagnosed when an older adult patient presents with the typical dermatomal vesicular rash and pain. During the rash phase, herpes simplex has the most similar presentation to herpes zoster. Evidence supporting herpes simplex includes a presentation in a younger population, multiple reoccurrences of lesions, and the absence of chronic pain. The lesions themselves are more difficult to distinguish, particularly when herpes zoster occurs in areas commonly affected by simplex (eg, oral, genital, buttock lesions). The differential diagnosis also includes contact dermatitis, burns, and vesicular lesions associated with fungal infections, but the history and examination usually make the distinction clear.

The diagnosis of herpes zoster may need laboratory diagnostic testing when differentiating herpes zoster from HSV infection is necessary, for suspected organ involvement, and for atypical presentations, particularly in the immunocompromised host. The preferred diagnostic test is PCR because it is highly sensitive and specific, and results are available within hours of specimen receipt. The best specimen is vesicle fluid, which contains abundant VZV. Lacking vesicle fluid, acceptable specimens include lesion scrapings, crusts, tissue biopsy, or cerebrospinal fluid.

MANAGEMENT

Acute Herpes Zoster

The principal goal of the treatment of herpes zoster in older adults is to decrease the length of the acute attack and to reduce or eliminate pain. Patients should be instructed to keep the rash clean and dry to decrease the chances of developing bacterial superinfection. Cool compresses, calamine lotion, cornstarch, or baking soda may help to reduce local symptoms and speed the drying of the vesicles.

Antiviral Therapy

The guanosine analogs acyclovir, famciclovir, and valacyclovir are phosphorylated by viral thymidine kinase to a triphosphate form that inhibits VZV DNA polymerase and VZV replication (**Table 107-1**). Valaciclovir and famciclovir are prodrugs, which have greater bioavailability, resulting in higher blood levels and a less frequent dosing schedule. Randomized controlled trials indicate that oral acyclovir (800 mg five times a day for 7 days), famciclovir (500 mg every 8 hours for 7 days), and valaciclovir (1 g three times a day for 7 days) reduce acute pain and the duration of chronic pain in older patients with herpes zoster who are treated with [illegible] of rash onset. The most common adverse effects are nausea, vomiting, diarrhea, and headache. The effectiveness of antiviral agents has not been evaluated for treatment after 72 hours have elapsed since the onset of rash. An exception to the 72-hour threshold for treatment is made for patients who have ophthalmic zoster or patients that continue to have new vesicle formation. Topical antiviral therapy is not an effective treatment regardless of when started.

Analgesics

Acute herpes zoster pain requires the use of standardized pain measures, scheduled analgesia, and close monitoring

TABLE 107-1 ■ ANTIVIRAL TREATMENT OF HERPES ZOSTER IN IMMUNOCOMPETENT AND IMMUNOCOMPROMISED PERSONS

IMMUNE STATUS	TREATMENT
Immunocompetent	
Age ≥ 50 years, and patients of any age with cranial nerve involvement (eg, ophthalmic zoster)	Famciclovir 500 mg PO every 8 hours for 7 days or Valacyclovir 1 g PO every 8 hours for 7 days or Acyclovir 800 mg PO five times a day for 7 days[a]
Immunocompromised	
Mild compromise	Famciclovir 500 mg PO every 8 hours for 7–10 days or Valacyclovir 1 g PO every 8 hours for 7–10 days or Acyclovir 800 mg PO five times a day for 7–10 days[a]
Severe compromise	Acyclovir 10mg/kg every 8 hours for 7–10 days[a]
Acyclovir resistant	Foscarnt 40 mg/kg IV every 8 hours until healed

[a]Adjust acyclovir dosing for renal function.

to adjust dosing and detect adverse effects. Opioid and nonopioid analgesia are a standard of care for the acute pain syndrome. The choice of treatment will depend on the patient's pain intensity, comorbidities, and preferences. Patients with mild pain may be managed with acetaminophen or nonsteroidal agents. Patients with moderate to severe pain usually require treatment with an opioid analgesic (eg, oxycodone).

Corticosteroids

The anti-inflammatory effects of corticosteroids have been postulated to reduce herpes zoster symptoms but there remains controversy over their use. Randomized, controlled clinical trials demonstrated that corticosteroids do not prevent PHN when compared to placebo or acyclovir. These findings, the risk of VZV dissemination, and the adverse effects of corticosteroids in older adults argue against the routine use of these agents in herpes zoster. However, clinical trials have shown reductions in acute pain and one randomized controlled trial showed benefits in improvement in time to uninterrupted sleep, return to routine activities, and cessation of analgesic medications among patients with no relative contraindications to corticosteroids. In the trial, prednisone was administered orally at 60 mg/day for days 1 to 7, 30 mg/day for days 8 to 14, and 15 mg/day for days 15 to 21. Therefore, corticosteroids may be useful in reducing moderate-to-severe pain and improving [illegible] antiviral agents and analgesics. Corticosteroids are used to treat VZV-induced facial paralysis and cranial polyneuritis to improve motor outcomes and provide pain relief. If prescribed, corticosteroids should always be used in conjunction with antiviral agents.

Adjuvant Agents

If moderate-to-severe herpes zoster pain is inadequately relieved by antiviral agents in combination with oral analgesic medications and/or corticosteroids, then other therapies to consider include gabapentin or pregabalin or neural blockade. The use of these anticonvulsant agents must be balanced against adverse effects of sedation, dizziness, ataxia, and peripheral edema. A randomized controlled trial of antiviral therapy, oral analgesics, and a single epidural block with bupivacaine and methylprednisolone compared to antiviral therapy and oral analgesics alone showed that neural blockade reduces acute pain but does not prevent PHN.

Ophthalmic zoster with ocular involvement requires ophthalmological consultation. Oral antiviral and analgesic therapy is the mainstay of treatment as it is for herpes zoster presenting on the rest of the body but other ocular therapies may be necessary. These treatments are best managed by an ophthalmologist and include antibiotic ophthalmic ointment to prevent bacterial infection of the ocular surface, topical steroids to reduce ocular inflammation, mydriatics as needed for iritis, and ocular pressure-lowering drugs as needed for glaucoma.

PREVENTION

Recombinant Zoster Vaccine

The adjuvanted recombinant zoster vaccine (RZV; Shingrix©) is a powerful tool to prevent herpes zoster and PHN in older adults. It contains two components: recombinant VZV surface glycoprotein E (gE) antigen and the AS01B adjuvant system. AS01B adjuvant contains monophosphoryl lipid A (MPL) and QS-21, a saponin, combined in a liposomal formulation. VZV gE elicits an immune response against VZV, while the adjuvant strongly potentiates the immune response by enhancing the innate immune response to the antigen.

Efficacy

The efficacy of RZV was demonstrated in over 30,000 immunocompetent community-dwelling elders in the randomized, blinded, placebo-controlled Zoster Efficacy Study in Adults 50 Years of Age or Older (ZOE-50) and Zoster Efficacy Study in Adults 70 Years of Age or Older (ZOE-70) (**Figure 107-4**). The intervention was two doses of RZV at a 2-month interval. The median follow-up time was 3.2 years in the ZOE-50 study and 3.7 years in the ZOE-70 study. The vaccine efficacy (VE) for the prevention of herpes zoster was 97% (95% confidence interval [CI] = 89.6–99.3) in persons aged 50–59 and 97% (95% CI = 90.1–99.7) in persons aged 60–69 (see **Figure 107-3**). The VE against herpes zoster was 91% (95% CI = 86.8–94.5) in participants aged 70 years or older using pooled data from the ZOE-50 and ZOE-70 studies.

VE in the first year after vaccination was 98% (95% CI = 90.9–99.8). It was 85% (95% CI = 69.0–93.4) or higher for the remaining 3 years of the study in persons aged 70 or older.

Safety and Reactogenicity

The only contraindication to RZV is a history of a severe allergic reaction to any component of the vaccine or after a previous dose of RZV. Serious adverse events, potential immune mediated diseases, and deaths were reported at similar rates in vaccine and placebo groups in the clinical trials.

Using pooled data from the ZOE-50 and ZOE-70 trial from all participants, the most common solicited adverse reactions after RZV administration were injection site pain (78%), myalgia (45%), fatigue (45%), headache (38%), shivering (27%), fever (21%), and gastrointestinal symptoms (17%). **Table 107-2** shows the frequency of adverse reactions by age group. The frequency of adverse reactions decreased with increasing age. Grade 3 reactions are defined as those that prevented normal activity. **Table 107-2** shows the frequency of grade 3 reactions by age group, which also decreased with increasing age. About 16% of participants 50 years and older experienced any grade 3 reaction.

To better characterize the impact of reactogenicity on functional status and quality of life in older adults, a separate study of 401 adults age 50 and older who received RZV

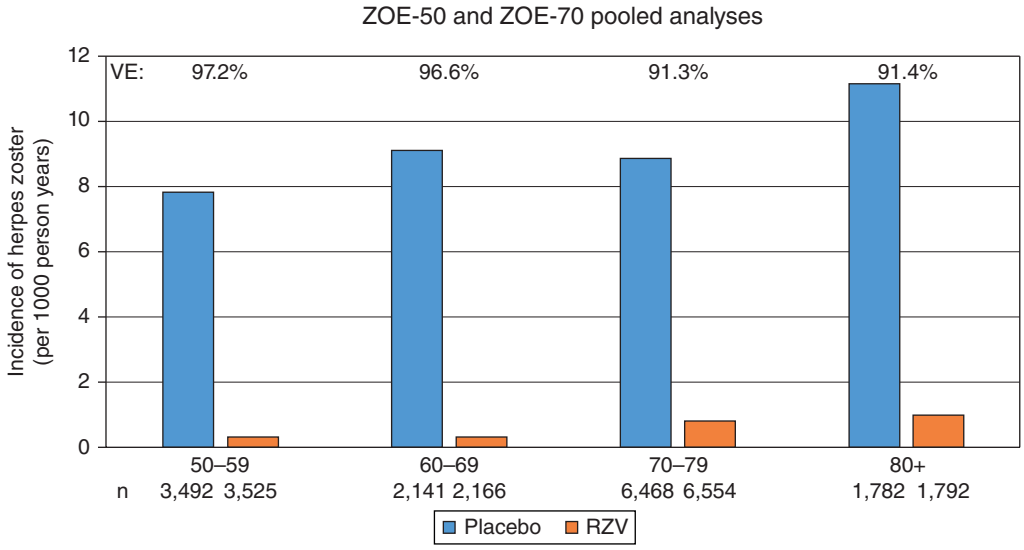

FIGURE 107-4. Recombinant zoster vaccine efficacy (VE) against herpes zoster from Zoster Efficacy 50 (ZOE-50) and Zoster Efficacy 70 (ZOE-70) pooled analyses. (Data from the US Food and Drug Administration. https://www.fda.gov/downloads/BiologicsBloodVaccines/Vaccines/ApprovedProducts/UCM581605.pdf.)

found that overall the physical functioning and quality of life of older adult were not affected by the first or second dose of RZV over 7 days postvaccination. However, participants who experienced a grade 3 reaction after either dose had a transient clinically important decrease in physical functioning and quality of life, affecting activities such as walking distances, carrying groceries, climbing stairs, lifting items, and moderate to vigorous activities. The impact was noted on days 1 to 2 post vaccination and resolved by day 3.

Given that frailty is strongly associated with adverse health outcomes, investigators studied how frailty impacts RZV VE, immunogenicity, reactogenicity, and safety. In secondary analyses of ZOE-50 and ZOE-70 studies, a baseline frailty index (FI) was created retrospectively following previously validated methods using preexisting comorbidities and patient reported outcomes. Participants were categorized as nonfrail (FI ≤ 0.08), prefrail (FI = 0.08–0.25), or frail (FI ≥ 0.25) for stratified analyses. RZV VE against HZ was consistently above 90% for all frailty categories (nonfrail: 96% [95% CI: 91.6–98.2], prefrail: 90% [84.4–94.4], frail: 90% [75.4–97.0]). The RZV group demonstrated robust antibody responses postdose 2 across frailty categories. In the RZV group, the percentage of participants reporting postvaccination reactions decreased with increasing frailty.

TABLE 107-2 ■ RZV LOCAL AND GENERAL ADVERSE EVENTS (%) BY AGE[a]

EVENT	50–59 YEARS	60–69 YEARS	≥ 70 YEARS
Injection site reaction, pain	89	83	69
Grade 3[b]	10	6.9	4.0
Myalgia	57	49	35
Grade 3	8.9	5.3	2.8
Fatigue	57	46	37
Grade 3	8.5	5.0	3.5
Headache	51	40	29
Grade 3	6.0	3.7	1.5
Shivering	36	30	20
Grade 3	6.8	4.5	2.2
Fever	28	24	14
Grade 3	0.4	0.5	0.1

[a]Reactogenicity subgroup within 7 days after vaccination, vaccine $n = 4884$.
[b]Grade 3—prevented normal activity.
Data from the US Food and Drug Administration. https://www.fda.gov/downloads/BiologicsBloodVaccines/Vaccines/ApprovedProducts/UCM581605.pdf.

Clinical Guidance

The Advisory Committee on Immunization Practices (ACIP) at the CDC recommends RZV for the prevention of herpes zoster and related complications in immunocompetent adults aged 50 or older including individuals who previously received zoster vaccine live (ZVL; Zostavax©). RZV is included in the routine adult immunization schedule for older adults.

The lyophilized VZV gE antigen is reconstituted with the liquid adjuvant to comprise the vaccine and administered as a 0.5 mL intramuscular (IM) injection. RZV is administered in a 2-dose series. Screening for a prior history of varicella or herpes zoster before administration of the first dose is not necessary. After the first dose, the second dose should be administered 2 to 6 months later. If the patient does not return within that time window, then the second dose should be administered at the next available opportunity. It is not necessary to restart the series. RZV may be coadministered with influenza, pneumococcal, and Tdap vaccines.

It is essential to counsel the patient about potential injection site and systemic reactions before administering both doses of RZV. They may feel pain, redness swelling at the injection site or muscle pain, tiredness, headache, chills, fever or nausea. About one in six persons have reactions that transiently prevent them from doing normal activities. Most symptoms resolve in 1 to 3 days. Patients should be counseled to avoid strenuous activities for a few days after vaccination. Reactions to the first dose do not strongly predict reactions to the second dose.

The ACIP provides no recommendation, either for or against, the use of RZV in immunocompromised populations. The FDA label does not list immunocompromised as a contraindication or precaution so administering RZV to immunocompromised individuals is not off-label prescribing to persons 50 years and older; it is off-label for persons younger than 50 years. Currently, best practice is to allow for RZV use in immunocompromised persons on a case-by-case basis if potential benefit is judged to outweigh the risk, using shared decision. Current information from randomized controlled trials indicate that RZV was immunogenic and had no safety concerns in small numbers (100–1000) of different immunocompromised populations including hematopoietic stem cell transplant (HSCT), kidney transplant, hematologic malignancies, solid cancers on chemotherapy, HIV, and immune-mediated disease. RZV VE was demonstrated prospectively in HSCT (VE = 68%) and in post-hoc analyses in hematologic malignancies (VE = 87%) and immune-mediated diseases (VE = 91%).

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Chapter 108

Influenza, COVID-19, and Other Respiratory Viruses

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INFLUENZA

Viruses are an important threat to the health of older adults. Each year, it is estimated that influenza alone is associated with the death of 36,000 older adults in the United States. Despite immunization, outbreaks of influenza occur regularly in nursing homes and other long-term care facilities. This section of the chapter summarizes the biological, epidemiological, and clinical features of influenza that are relevant to older adults, with a particular emphasis on prevention.

Clinically Relevant Biological Characteristics of Influenza Virus

To understand the impact of influenza in older adults, it is important to be familiar with several key characteristics of the virus. The structure of influenza consists of an envelope with a central nucleic acid core composed of single-stranded RNA that encodes key structural proteins of the virus including hemagglutinin and neuraminidase. There are three types of influenza viruses: A, B, and C. However, only A and B are clinically relevant. Influenza A viruses are characterized by the structure of the hemagglutinin, a surface protein that binds glycoprotein on the surface of respiratory epithelial cells, allowing the virus to enter by forming an endosome and then using the protein-making machinery of the cell to replicate itself. Each year, new mutations occur resulting in small changes in the hemagglutinin ("antigenic drift") hence the reason why influenza vaccine needs to be reformulated and given annually. The other surface projection, neuraminidase, cleaves terminal sialic acid residues from carbohydrate moieties on surfaces of infected cells, promoting the release of virions that go on to infect other cells. As discussed below, this is a key target for neuraminidase inhibitors, thus preventing the influenza virus from replicating.

An important feature of influenza is the segmented structure of the RNA at the core of the virus, with each of eight segments coding for a structural or enzymatic component of the virus. This gives the virus the potential to recombine with influenza viruses of animal origin, forming a virus with a novel genotype and hemagglutinin to which there is no preexisting immunity. This is known as "antigenic shift" and was responsible for pandemics in

Learning Objectives

- Understand the biology of influenza virus that leads to "drift" and "shift" in surface protein structure and their impact on vaccine formulation and the emergence of influenza pandemics.
- Recognize clinical situations where there is a need for diagnostic testing to confirm a specific viral etiology in symptomatic older adults and recommend the specific tests appropriate in different clinical scenarios.
- Identify appropriate interventions to control outbreaks of influenza, respiratory syncytial virus (RSV), and severe acute respiratory syndrome coronavirus-2 (SARS-CoV-2) in a nursing home.
- Recognize the appropriate use of antiviral therapies for influenza in seniors.
- Review the biological and other features of SARS-CoV-2 infection that contribute to pandemic spread.

Key Clinical Points

1. **The influenza virus has a segmented RNA core that encodes key proteins on the viral surface including hemagglutinin and neuraminidase. Annual mutations in the RNA lead to "drift" in these proteins requiring annual reformulation and administration of the vaccine.**
2. **When entire segments of the influenza RNA recombine in animals, it can lead to an antigenic "shift" and may result in a pandemic as there may be little or no immunity in the population.**
3. **Detection of influenza virus by direct antigen testing is insensitive in older adults, and a negative test does not rule out influenza infection. Reverse transcriptase polymerase chain reaction (RT-PCR) is more than 90% sensitive. Similarly, RSV is best detected by RT-PCR.**

(Continued)

(Cont.)

4. **RSV is clinically indistinguishable from influenza in older adults—both may present subtly or primarily as an exacerbation of underlying chronic illnesses.**
5. **Antiviral therapy with neuraminidase inhibitors is indicated for treatment of influenza and as chemoprophylaxis for nursing home outbreaks. Supportive care is currently the only treatment for RSV.**
6. **Influenza immunization is recommended for all older adults. Enhanced influenza vaccines are more effective than a standard-dose vaccine in seniors.**
7. **Immunization of staff reduces influenza mortality for nursing home residents.**
8. **Transmission of SARS-CoV-2 is highest at the beginning of infection, often prior to the onset of symptoms.**
9. **Symptomatic cases of SARS-CoV-2 present with fever, cough, fatigue, and dyspnea, though older adults may have atypical presentations.**
10. **Older adults, particularly those residing in nursing homes, are at high risk of complications, hospitalization, and death as a result of influenza, RSV, and SARS-CoV-2 infections.**

1957–1958, 1968–1969, and 2009. Of note, the highest mortality rates occurring in the 1918–1919 and the 2009 pandemics were not in older adults but in younger adults, perhaps due to cross protection from the many influenza strains to which older adults were exposed over a lifetime. However, mortality in interpandemic periods is highest in those at the extremes of age. At present, there are only three hemagglutinins (H1, H2, and H3) and two neuraminidases (N1 and N2) that have developed a stable lineage in humans.

General Epidemiology of Influenza

Epidemics of influenza occur annually between November and April in the Northern Hemisphere. In any given season, there may be several strains circulating, hence the annual influenza vaccine contains multiple influenza strains.

Important Host Considerations in Older Adults

During the interpandemic period, older adults and particularly residents of long-term care facilities are equally likely to contract influenza as young, healthy adults, but are among those at highest risk for complications of influenza. Rates of hospitalization for influenza in older persons range from 136 to 508 per 100,000 persons versus 10 to 25 for those 5 to 49 years of age. The presence of chronic conditions, such as chronic lung disease, congestive heart failure, conditions that predispose to aspiration, and metabolic disease, increases the risk for complications following infection with influenza. Many of these conditions occur predominately in older age groups. Moreover, studies that have assessed complications have found that age older than 65 is independently associated with increased risk for influenza complications.

Based on our current understanding, the causes of death as a result of influenza in older adults in the interpandemic period include viral pneumonia, a bacterial infection complicating the influenza infection, myocardial infarction, stroke, and exacerbation of underlying comorbid conditions. Infection with influenza virus predisposes to *Streptococcus pneumoniae* or *Staphylococcus aureus* infection, which most often results in bacterial pneumonia. Deficits in innate immunity (phagocytes, natural killer cells) and acquired immunity (T-cell function, cytokine activity, antibody response) are all felt to play a role, as discussed in Chapter 3.

Clinical Presentation

Young, healthy individuals with influenza characteristically present with a sudden onset of fever, cough, myalgia, sore throat, and headache. Fever and cough have been shown in a systematic review to be the best predictors of influenza in the general population. In older adults, however, the presentation may be more subtle with cough and change in baseline temperature predominating. Often, older adults present with worsening of comorbid conditions such as a chronic obstructive pulmonary disease or congestive heart failure exacerbation. One of the most important factors in making the diagnosis of influenza, whether on clinical grounds or through diagnostic testing, is the local influenza activity. That is, if a community is experiencing an outbreak of influenza, particularly if the incidence of influenza is at its peak, fever and cough in an older person increase the likelihood of infection with influenza. In the nursing home setting, it is essential to obtain prompt diagnostic testing because the clinical presentation is nonspecific. Even when there are peaks of influenza in the community, RSV or other respiratory viruses can circulate. Additionally, an outbreak of influenza in a nursing home warrants chemoprophylaxis (see further discussion) as well as immunization of nonimmunized residents in whom it is safe to do so.

Specific Diagnostic Tests

The traditional method to detect influenza has been viral culture, which is limited by moderate diagnostic accuracy and delay in obtaining results (**Table 108-1**). Rapid tests include direct fluorescent antigen (DFA), or direct immunofluorescence, where monoclonal antibodies labeled with fluorescent material are directed to influenza cell coat antigens. A result can be reported within hours. Rapid enzyme-linked immunosorbent assay (ELISA) tests are

TABLE 108-1 ■ SUMMARY OF TEST CHARACTERISTICS OF DIFFERENT DIAGNOSTIC TESTS FOR VARIOUS RESPIRATORY VIRAL PATHOGENS IN OLDER ADULTS

	CULTURE	RAPID EIA	DFA/IFA	PCR	SEROLOGY
Influenza A	++	+	+	+++	++[a]
Influenza B	++	+	+	+++	++[a]
RSV	+	+/–	+/–	+++	+++

+/–, available but poor sensitivity; +, fair sensitivity; ++, good sensitivity; +++, optimal sensitivity.
[a]Interpretation complicated by vaccination.

commercially available but are very insensitive in older adults, with sensitivities as low as 20%. Nucleic acid amplification testing, such as RT-PCR, is the most promising test since it is over 95% sensitive and specific and the turnaround time can be rapid. The drawback, however, is that not all laboratories are capable of performing PCR daily.

Infection Control Aspects

The incubation period of influenza, the period of time from initial infection to development of symptoms, is typically 2 days but can range from 1 to 4 days. An infected individual becomes contagious 1 day prior to the onset of symptoms. In adults, viral shedding at levels high enough to cause transmission occurs over 5 or 6 days. The majority of spread of influenza is large droplet caused by coughing and sneezing and hence medical workers should wear a mask to prevent infection and spread. In terms of practical implications, use of "respiratory etiquette," coughing, and sneezing into tissues, for example, may help prevent spread to older adults who are susceptible to complications, although supporting epidemiologic evidence is limited.

Immunization

Many influenza vaccine formulations are now available in the United States. The most well-known influenza vaccine is the standard-dose quadrivalent inactivated vaccine, abbreviated as SD-IIV4, which contains components of an H1N1, an H3N2, and two B strains.

For the same reasons that older adults are more likely to have complications due to influenza infection, older adults respond less well to influenza immunizations. A 2014 systematic review summarized 35 influenza vaccine effectiveness studies performed in 15 countries. Cases were influenza positive; controls were patients who also had an acute respiratory illness but who tested negative for influenza. Vaccine effectiveness for the prevention of medically attended acute respiratory illness was found to be 38% to 70% during seasons of vaccine and circulating strain match and 15% to 59% during seasons of mismatch. These data show that influenza immunization plays an important but modest role in prevention.

Due to decreased immune responses and poorer clinical responses when SD-IIV4 is administered in seniors (compared with young adults), enhanced influenza vaccines have been developed and approved for use. A high dose vaccine quadrivalent formulation (HD-IIV4) has four times the dose of each component versus the standard dose. This high dose formulation has been shown to generate higher antibody levels in older adults than SD-IIV4 and is more effective for the prevention of illness and hospitalization due to influenza. The MF59 adjuvanted influenza vaccine, abbreviated aIIV4, also demonstrates increased immunogenicity and effectiveness in older adults compared to the SD-IIV4. Observational studies show MF59 adjuvanted influenza vaccines are associated with reduced hospitalization risk for not only influenza, but also pneumonia and cardiovascular diagnoses—common complications of influenza infection. The quadrivalent, recombinant influenza vaccine (RIV4) also shows promise in preventing intensive lifestyle intervention in older adults. At this time, the Advisory Committee on Immunization Practices recommends the use of any of these vaccines in older adults.

Influenza and Long-Term Care

Residents of nursing homes and other long-term care facilities are at high risk for complications of influenza, including pneumonia, hospitalization, and death. This is likely because there is close contact between residents and staff in these facilities, increasing the exposure to and transmission of influenza, and because the majority of residents have comorbidities, including cognitive impairment, chronic lung disease, and stroke. Outbreaks of influenza are well characterized in nursing homes, often described as "explosive" in that the onset is relatively abrupt. Such outbreaks are usually first detected by nursing home staff who notice a higher incidence of respiratory symptoms than usual on specific units, although at least one surveillance study has shown that such outbreaks are commonly missed even when active surveillance by trained nurses is performed. Such outbreaks are best managed using a multifaceted approach that includes the following: cohorting residents, increasing adherence to hand hygiene, chemoprophylaxis with antiviral agents, and immunization of those not previously vaccinated. It should be noted that because of space and staffing limitations, it is usually not feasible to cohort ill residents together as a means of separating them from noninfected residents in long-term care facilities. Administering neuraminidase inhibitors as chemoprophylaxis to noninfected residents helps to reduce further spread. In one randomized controlled trial where long-term units were randomized to zanamivir or to rimantadine, the risk of influenza was 3% in the zanamivir arm versus 8% in the rimantadine arm ($p = 0.038$). More recent data suggest that immunizing nursing home staff against influenza benefits residents. There have been four cluster-randomized controlled trials showing that immunizing health care workers against influenza reduces mortality in residents of long-term care facilities. Vaccinating nursing home staff against influenza may prevent influenza-like illness, use of health services, and deaths in nursing home residents, even more so than immunizing residents.

Use of Antivirals to Prevent and Treat Influenza in Older Adults in the Community

There are six antivirals that are approved for treating influenza: amantadine (oral), rimantadine (oral), zanamivir (inhaled), oseltamivir (oral), peramivir (IV), and baloxavir marboxil (oral). Amantadine and rimantadine are only active against influenza A, and currently circulating influenza A strains are predominantly resistant to both; their use is no longer recommended. The neuraminidase inhibitors, zanamivir and oseltamivir, are active against influenza A and B. They are 70% to 90% effective for prophylaxis in preventing influenza and when used as treatment to reduce clinical severity if given within 48 hours of symptom onset. However, the feasibility of using these agents for treatment is reduced since the majority of patients in the community present over 48 hours following symptom onset. It is important to note that the doses of treatment and prophylaxis are different; hence, it is important to use treatment doses if a patient is symptomatic. Zanamivir and oseltamivir are associated with few adverse effects. Zanamivir may worsen or provoke respiratory distress in those with underlying obstructive disease. Oseltamivir should be dosed based on renal function. In early 2015, the Food and Drug Administration (FDA) approved a third neuraminidase inhibitor, peramivir, for the treatment of influenza that is only available in an IV formulation. Baloxavir marboxil, a polymerase inhibitor, was approved in 2018 for treatment of uncomplicated cases of both influenza A and B. Similar to the neuraminidase inhibitors, it must be used within 48 hours of symptom onset. Unfortunately, viruses develop resistance to baloxavir very quickly, limiting its use.

Indirect Benefit of Immunization in Children to Protect Older Adults

There is observational data to suggest that children play an important role in the spread of influenza in the community. For example, data from a 3-year longitudinal surveillance study of children and adults in New York State demonstrated that children were about twice as likely to acquire and shed influenza compared to adults. In a longitudinal study conducted in Seattle from 1968 to 1974, elementary and junior high school students had the highest rates of influenza during epidemics, reaching 54%. In the Tecumseh, Michigan, studies from 1976 to 1981, over one-third of children between the ages of 5 and 14 had influenza virus isolated from specimens, and the highest rate was among persons with febrile respiratory illness. Serological data revealed similar results; from 1977 to 1978, children in 5 to 9, 10 to 14, and 15 to 19 age groups had an approximately 30% infection rate with H3N2. This was over twice the rate seen in adults. There were similar infection rates with influenza B in children aged 5 to 14, rates 14-fold higher than those in adults older than 20 years.

Given that school-aged children appear to play an important role in introducing and transmitting influenza into households and hence into the community, immunizing these children may interrupt spread to older adults who are at high risk for complications. In the Tecumseh studies, selectively immunizing 86% of children in this 7500-person community with inactivated influenza vaccine reduced influenza in adults by a third when compared to an adjacent community where children were not immunized. More evidence for the potential benefit of immunizing children with inactivated vaccine has been derived from an analysis of the effect of influenza vaccination in Japan. Japan began a program of immunizing school-aged children in 1962 and continued this policy until 1994. The effect of this policy was to dramatically reduce excess mortality rates to values similar to those in the United States. The fact that there was a rapid increase in excess deaths after 1994, the year in which mass immunization formally ended, supports the conclusion that the effects observed in earlier years were because of vaccine-induced herd immunity, although it is possible that social factors may have amplified the effects of this program. The authors explain their findings by hypothesizing that the high levels of vaccination in school children protected transmission of influenza to their grandparents. There have been several more recent observational studies where children have been immunized showing a reduced attack rate of influenza in adults. However, randomized clinical trial evidence of immunizing children to prevent influenza in older adults is limited.

RESPIRATORY SYNCYTIAL VIRUS

RSV has been long recognized as a major cause of bronchiolitis in children. Since the 1970s, there have been numerous reports about the burden of RSV in older adults, particularly those residing in nursing homes. RSV is now recognized as an important contributor to morbidity in both community-dwelling older adults as well as in residents of long-term care facilities. This section will describe the epidemiology of RSV in both settings including clinical, diagnostic, and management issues.

Characteristics of Respiratory Syncytial Virus

RSV is an enveloped RNA virus that can be classified into two major groups, A and B, based on antigenic and genetic properties. An important feature of RSV is that reinfections occur frequently throughout life and into older age. Because RSV infection does not produce robust immunity, developing a vaccine has been a major challenge. In addition, RSV is a very labile virus, meaning it can be difficult to grow if it is not transported carefully, making traditional diagnostic methods such as culture and antigen less sensitive outside of the hospital setting.

General Epidemiology

Like influenza, RSV circulates during winter months in temperate climates roughly beginning in October and lasting until early spring. Often, clinical cases precede

influenza epidemics. The incubation period of RSV is 3 to 5 days. The majority of those who are infected show respiratory symptoms. In contrast to infants who shed higher titers (10^{5-6} plaque-forming units/mL of secretion) for longer time periods (10–14 days), adults tend to shed lower titers of the virus (10^{2-3} plaque-forming units/mL of secretion for 3–6 days). RSV is spread primarily by large droplets and fomites.

Community-Dwelling Older Adults and Respiratory Syncytial Virus

The best data about RSV in older persons living in the community are derived from a cohort study conducted in Rochester, New York, from 1999 to 2003. This cohort was composed of 608 healthy people aged 65 and older. Forty-six (8%) of the 608 participants developed RSV, compared to 56 (10%) of high-risk older participants (defined as having congestive heart failure or chronic pulmonary disease). The impact of RSV infection was important in both the healthy older adults group and in the high-risk group: 17% and 23%, respectively, of these groups made physician office visits. Morbidity was greater in the high-risk group, where 16% required hospitalization. The study was also useful because it allowed for a comparison to influenza: 42% of healthy older participants and 60% of high-risk participants required office visits for influenza, rates statistically significantly higher than for RSV. Since then, RSV has been shown in other studies to cause similar if not higher hospitalization rates than influenza in older adults.

Epidemiology of Respiratory Syncytial Virus in Long-Term Care Facilities

RSV attack rates during nursing home outbreaks have been reported to be variable, ranging from 2% to 90%. However, most involved relatively small numbers of residents and included the use of diagnostic assays with varying levels of sensitivity. In one prospective study conducted in Rochester, where a sensitive enzyme immunoassay was used, the overall attack rate was 7% and the virus was identified as a cause of 27% of illness. One retrospective cohort study estimated the burden of illness in nursing homes in Tennessee by linking cardiopulmonary hospitalization, medical utilization, and death to viral activity in the general community during a period of 88,581 person-years. According to these data, RSV accounted for 15 hospitalizations and 17 deaths per 1000 residents.

Clinical Features of Respiratory Syncytial Virus in Older Adults

The clinical features of RSV cannot definitively be distinguished from those of other viruses, though fever is typically less pronounced than for influenza and wheezing more pronounced. The nonspecific nature of the clinical presentation indicates the need for diagnostic testing. Signs and symptoms include rhinorrhea, nasal congestion, sore throat, cough, and dyspnea. Similar to patients infected with influenza, patients with RSV may present with exacerbation of underlying conditions.

Diagnosis

This can be done by viral culture, antigen detection in respiratory secretions, nucleic acid amplification, or serology. For diagnosis in real time, only the first three tests are appropriate. It is important to note that viral culture is only 50% sensitive. Similarly, detection of RSV antigens using immunofluorescence is very insensitive in older adults. In contrast, RT-PCR is relatively sensitive and specific. As the use of multiplex PCR systems increases, RSV will likely be recognized more often as the causative agent in the clinical setting.

Therapy

Care is mainly supportive in the older adult with RSV infection. The only licensed antiviral with activity versus RSV is ribavirin, but there are sparse data on which to base treatment recommendations in adults. Practical considerations surrounding ribavirin include that the administration of this aerosolized drug to older adults using face masks may be difficult, and that ribavirin is teratogenic, which is important for the health care provider who is delivering care and the family that is entering the patient's room.

Prevention

Transmission of RSV is believed to be caused by large droplets and fomites. Strategies such as increasing hand hygiene, particularly in nursing homes, may help to reduce spread. Isolation practices for RSV include gowning and gloving when taking care of a patient with RSV. At the present time, there are no licensed RSV vaccines.

SARS-COV-2 AND OTHER CORONAVIRUSES

Coronaviruses are a large family of enveloped, positive-strand RNA viruses that are endemic in animals and typically cause mild upper respiratory symptoms in humans, known as "the common cold." In the past two decades, however, three novel coronaviruses have transitioned from their animal hosts to cause severe disease in humans: SARS-CoV in 2002, Middle East respiratory syndrome (MERS-CoV) in 2012, and SARS-CoV-2 in 2019, the cause of pandemic coronavirus disease known as COVID-19. The COVID-19 pandemic has had global impact unparalleled in this generation, and like influenza, disproportionately impacts older adults. This section will review SARS-CoV-2 transmission, clinical manifestations, diagnosis, and infection control measures.

Characteristics of Coronaviruses and SARS-CoV-2

Coronaviruses were named for their spike-like surface projections that resemble a crown. Bats are an important

ecological host of coronaviruses and likely serve as reservoirs. Zoonotic transmission—the spread of pathogens from animals to humans—leads to human disease. The virus must first acquire mutations that allow it to begin replicating in human cells. These mutations typically arise randomly in the natural reservoir. Transmission of the virus to a human host can occur directly, via saliva, bites, or through the air, or by way of an intermediate species. Intermediate hosts—palm civets for SARS-CoV and dromedary camels for MERS-CoV—contributed to zoonotic transmission of these viruses. There is no conclusive evidence regarding when and where SARS-CoV-2 transitioned to humans. Bats and pangolins have been linked to SARS-CoV-2 as possible zoonotic reservoirs, and it is not known whether an intermediate host was involved. Adaptation to a human host is necessary but not sufficient to cause disease outbreak; human-to-human transmission must also occur.

Transmission of SARS-CoV-2

Similar to SARS-CoV, SARS-CoV-2 binds to the angiotensin-converting enzyme 2 receptor to enter human respiratory epithelial cells and replicates rapidly throughout the respiratory tract. While SARS-CoV and MERS largely reproduce in the lower respiratory tract, high viral loads of SARS-CoV-2 are found in the upper respiratory tract. Though SARS-CoV and influenza transmission occur largely after symptoms begin and peak with severity of disease, pharyngeal shedding of SARS-CoV-2 is highest at the beginning of infection. This means transmission may occur prior to the onset of symptoms and before a person knows they are infected. Transmission through asymptomatic or presymptomatic cases is thought to be a major contributor to pandemic spread, estimated in one study as accounting for up to 79% of documented cases. Its tropism for the upper respiratory tract and asymptomatic spread makes SARS-CoV-2 a highly transmissible pathogen. Case detection is challenging and spread difficult to prevent without widespread public health measures.

The main mechanism of transmission is via respiratory droplets and airborne particles that are directly inhaled while in close contact with an infected person. It is hypothesized that some people, deemed "super spreaders," shed more infectious particles in speech and respiratory droplets than others, leading to high numbers of new infections related to large gatherings. Evidence for transmission via droplet contact with the conjunctiva has led to the recommendation for eye protection for all those caring for patients infected with SARS-CoV-2.

Coronaviruses have higher stability than the enveloped influenza virus, including stability on surfaces, and fomite transmission of SARS-CoV-2 is possible. Viable virus has been demonstrated to persist on inanimate surfaces for several days; however, the inoculum required to spread from touching surfaces is not known. Fomites are thus thought to be a possible, though not major source of viral transmission. Though prolonged viral shedding in fecal samples has been demonstrated, there is no evidence for fecal-oral spread of disease at this time.

Clinical Features of SARS-CoV-2

Coronaviruses have a longer incubation period—time from infection to symptom onset—compared to influenza. Various estimates of the incubation period of SARS-CoV-2 have been proposed. Most recent data estimates symptom onset at around 5 days after infection, with most people demonstrating symptoms within 12 days. There is evidence that older adults experience a longer incubation period, with symptoms typically occurring at 7 days after infection.

Clinical manifestations of COVID-19 disease range from asymptomatic or mild symptoms in the majority of cases, to severe respiratory failure and multiorgan dysfunction, which occur in a small proportion of infections. For those who develop symptoms, fever, cough, fatigue, and dyspnea are most common. However, atypical presentations are also common, particularly in older adults. One study found that 40% of patients with a mean age of 81 presented with atypical symptoms such as falls, generalized weakness, and delirium. A wide variety of symptoms, including myalgias, headache, diarrhea, nausea, vomiting, sore throat, and loss of sense of smell and/or taste have been associated with SARS-CoV-2 infection. Other notable clinical features include lymphopenia and other cytopenias, elevated inflammatory markers, and ground-glass opacities on chest imaging in those who develop respiratory symptoms.

Though the majority of cases are mild, the proportion of cases requiring respiratory support is higher than that of 2009 pandemic influenza. It is hypothesized that the rapid replication of SARS-CoV-2 in the respiratory tract can lead to recruitment of inappropriately high levels of pro-inflammatory cytokines and cells, referred to as a "cytokine storm." This excessive inflammatory response is thought to underlie more severe cases of SARS-CoV-2 infection.

Epidemiology of SARS-CoV-2

All ages are susceptible to SARS-CoV-2 infection. While early in the pandemic COVID-19 incidence was highest in adults age 50 and older, the median age of SARS-CoV-2 infection has decreased as testing has become more available to those with asymptomatic or mild cases of infection, who tend to be younger. While all ages are susceptible, severity of disease increases with increasing age. Severe illness occurs in a minority of patients, with the highest rate in those older than 70 years, and is rare in children. Variable case fatality rates have been reported, largely due to differences in testing; however, fatal illness occurs primarily in those older than 50 years, and up to 80% of deaths from SARS-CoV-2 have been in adults 65 years and older.

Frailty has been proposed as a means to capture the heterogeneity of older adults with SARS-CoV-2 infection and guide decisions regarding prognosis and treatment. One observational cohort study demonstrated an

association between frailty, measured using the clinical frailty scale, and 7-day mortality in adults hospitalized with SARS-CoV-2 infection. In this study, frailty was a better predictor of disease outcome than age or comorbidity alone. In addition to frailty, many common comorbidities are associated with increased risk of severe illness, progression to acute respiratory distress syndrome, and death, including type 2 diabetes, chronic obstructive pulmonary disease, heart failure, chronic kidney disease, obesity, cancer, and solid organ transplant.

SARS-CoV-2 is notable for its ability to cause outbreaks in congregate settings. Early outbreaks were reported on several cruise chips and later affected institutions like meat-packing plants, where employees work in close quarters, and college dormitories. Nursing homes, however, have been most critically affected by the COVID-19 pandemic. The combination of the close quarters of congregate living, a frail population with high personal care needs, and the conditions leading to chronic understaffing and lack of resources for nursing homes created a setting ripe for uncontrolled spread of infection and high mortality. Over a quarter of deaths have occurred in the postacute and long-term care sector, despite accounting for only 2% of the total confirmed cases as of November 2020.

The impact of physical distancing, loneliness, and social isolation on behavioral health and mortality, particularly for older adults and those who reside in nursing homes, has yet to be fully expounded. In one pre-pandemic study, social isolation, loneliness, and living alone corresponded to an average of 29%, 26%, and 32% increased likelihood of mortality, respectively, in older adults. This is comparable to the risk of unhealthy diet, physical inactivity, alcohol misuse, and smoking. The loss of social interaction may have particular negative consequences for those with dementia.

Though a smaller proportion of people infected with SARS-CoV-2 require hospitalization compared to those with SARS-CoV, as well as to those infected with 2009 pandemic influenza, the overall number of people infected is much higher. This has placed strain on health system resources, at times exceeding hospital and ICU capacity and leading to higher mortality in many areas of the world.

Diagnosis

RT-PCR for detection of viral RNA in a sample from the nasopharynx is the cornerstone of testing for active SARS-CoV-2 infection. Compared to other diagnostic tests, it has the highest sensitivity at early stages of infection, and specificity is also high; however, appropriate sample collection is vital. Turnaround time ranges from 15 minutes to several days. RT-PCR is time- and labor-intensive and requires a well-equipped laboratory, which also impacts turnaround time, particularly for facilities like nursing homes and rural hospitals that must send tests to a central laboratory.

Development of point-of-care (POC) testing has been a priority during the pandemic in order to provide rapid results that can guide clinical management, allocation of personal protective equipment (PPE), and activation of isolation protocols. Currently available POC devices automate multiple steps of lab-based immunoassay methods to detect viral antigens in nasal specimens in under an hour. Sensitivity is lower than RT-PCR, particularly early in the course of infection, with high specificity; however, these devices have been tested and received authorization for use on symptomatic persons only, and test characteristics for screening and testing asymptomatic persons are unknown.

While there is no role for serologic testing for antibodies to SARS-CoV-2 in diagnosing active infection, the immunology of SARS-CoV-2, including the course of antibody production and protection against future infection, is an active area of study.

Treatment

The global impact of the COVID-19 pandemic has stimulated an exceptional period of research and development of not only diagnostics but also a myriad of potential therapeutics. Several medications have shown benefit in specific populations—for example, the steroid dexamethasone reduces mortality in hospitalized patients requiring oxygen or mechanical ventilation, though confers no benefit in those not requiring respiratory support. Though several have been approved for emergency use in the United States, antivirals and monoclonal antibodies, as well as convalescent plasma, have yet to be evaluated in phase 3 trials; availability is also limited, and expense high, for these treatments. Hundreds of drugs continue in development or clinical trials; however, no treatment has demonstrated general prophylactic or therapeutic benefit against SARS-CoV-2.

Infection Control Aspects of the COVID-19 Pandemic

Without broadly effective therapeutics, infection control measures are paramount in mitigating the adverse health outcomes of the pandemic. Basic measures including hand hygiene and masks to reduce the spread of respiratory droplets, identical to those recommended for other respiratory viruses including influenza, have been demonstrated to reduce transmission. Cities and states across the United States also implemented restrictions on public gatherings to encourage social distancing and limit the spread of infection.

As discussed in a previous section, rapid, accurate, and readily available diagnostic tests are essential to identifying cases and isolating infected patients. SARS-CoV was mitigated in part due to rigorous contact tracing and case isolation measures, and countries able to implement similarly robust public health measures were able to rapidly control the spread of SARS-CoV-2. Gaps in diagnostics exist, most notably accurate tests for screening asymptomatic persons to identify those in the incubation phase of infection. Detection of these cases is critical to control asymptomatic and presymptomatic transmission.

Nursing home best practices have developed to include regular testing of staff and residents along with appropriate PPE and hand hygiene. Increases in nursing home cases and deaths are associated with increases of cases in the community, and staff testing frequencies are based on community positivity rates. Contract tracing and isolation of positive residents, as well as cohorting staff so that different groups care for positive and negative residents, are ideal but often unattainable due to staffing and resource limitations in this setting.

In addition to limiting spread of infection, a crucial approach to pandemic mitigation is to reduce the number of susceptible people via immunization. Promising vaccine candidates have demonstrated more than 90% effectiveness in preventing infection, and immunization can also have a role in reducing transmission and attenuating illness. Large-scale production and global vaccine deployment remain an ongoing challenge in combating the COVID-19 pandemic.

Concerns about emerging coronaviruses have been growing even prior to the COVID-19 pandemic. The global experience of encountering a novel virus with pandemic spread and an impact far beyond health care has highlighted the need for ongoing research into emerging infectious diseases. The need for coordinated international response and cooperation, not only in combating the COVID-19 pandemic but also in preparing for future pandemics, has become more evident than ever.

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